

Research paper

Optimal energy management strategy for electric vehicles and electricity distribution system based on quantile deep learning with enhanced African vulture optimization

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ABSTRACT

The increasing fluctuation in electricity prices due to unpredictable demand loads on the power grid has underscored the importance of finding optimal scheduling solutions for electric vehicles (EVs). Integrating EVs into the distribution system (DS) is critical for efficiently managing their energy needs. However, current EV scheduling approaches face challenges. Centralized solutions are often unreliable and complex, while existing decentralized methods struggle to efficiently schedule EVs in large networks with smart energy DS. To address these challenges, we propose a hybrid technique called Quantile Deep Learning with Enhanced African Vulture Optimization (Quantum VultureNet) for energy management systems (EMS) involving EVs and DS. This approach combines the African Vultures Optimization algorithm with elite mutation, learning based on dynamic opposition, and chaotic map-based population initialization. By enhancing exploration and exploitation capabilities, this hybrid method aims to improve the scheduling of EVs, particularly in dynamic environments. The primary goal of the Quantum VultureNet technique is to minimize system costs and power losses while optimizing energy management. The proposed system considers factors such as bidirectional energy trading, EV fleet arrival times in driving plans, Photo Voltaic uncertainty impacts on energy management systems, and factors affecting energy selling back to the grid. The model was implemented in MATLAB and compared with conventional methods. Simulation results demonstrate that the Quantum VultureNet technique effectively finds near-global optimum solutions with reduced computational complexity. This indicates the effectiveness of the proposed approach for EMS in EVs integrated with DS.

1. Introduction

Recently, vehicle electrification garnered significant heed and has quickly developed due to its potential to reduce environmental pollution and address ecological issues (Du et al., 2021a). EVs Charging Stations (EVCs) plays the crucial part in rise of energy, providing platforms for energy interaction between power grid and EV (Khan et al., 2018; Mythili et al., 2020). Additionally, the charging demand of EVs impact temporal and spatial supply of electric charge, highlighting the importance of efficient energy management(EM) among the EVs Charging Stations(EVCs), power grids along with EVs. Many researchers had focused on coordinating energy management(EM) between power grid and EVs (Wu and Sioshansi, 2017; Shajin and Rajesh, 2020).

Distribution system plays a crucial role in integrating EVs and managing their energy needs (Wu et al., 2016a; Thota et al., 2020a). An energy management(EM)method suggests that EV should contribute the operation efficiency of the distribution system by adjusting their charging plan accordingly (Ji et al., 2016a). Additionally, multistage EVs load optimization systems based on mixed linear program has been made to reduce distribution system transmission loss (Liu et al., 2019a). Also, an organized operation method aims in enabling merging of conventional energy and EVs transforming distribution systems(DS)into two-layer systems, with one layer serving traditional load and the other interacting with renewable energy and EVs (Abbasi et al., 2019a). As the use of EVs and EVCs grows, there is a need for coordinating power management between the distribution system and EVCs. For instance, a conception model was used to estimate total loads of EVCs integrated

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Nomenclature			
Abbreviation description			
EVs	Electric vehicles	FC	Fuel cell
DS	Distribution system	BU	Battery units
EMS	Energy management system	A-ECMS	Adaptive energy management strategy
EAVO	Enhanced African Vulture Optimization	BOA	Butterfly optimization algorithm
PV	Photovoltaic	PHEV	Plug-in hybrid electric vehicle
DOL	Dynamic opposition-based learning	RF	Random forest
EVCS	Electric Vehicle Charging Station	SVM	Support vector machine
HEMS	Home energy management system	WOA	Whale optimization algorithm
RES	Renewable Energy Sources	DRL	Deep reinforcement learning
FIT	Feed-in-Tariff	SHEV	Series hybrid electric vehicle
V2H	Vehicle-to-Home	HER	Heuristic experience replay
GEMS	Grid energy management system	SHETV	Series hybrid electric tracked vehicle
DP	Dynamic programming	DQL	Deep Q-learning
PMP	Pontryagin's minimum principle	HESS	Hybrid energy storage system
MPC	Model predictive control	DFLC	Dual fuzzy logical controller
ECMS	Equivalent consumption minimization strategy	NAG	Nesterov accelerated gradient
FFA	Fertile Field algorithm	ICE	Internal Combustion Engine
GBDT	Gradient Boost decision tree	EM	Electric Machine
FCHEV	fuel cell /ultra-capacitor hybrid electric vehicle	UDDS	Urban Dynamometer Driving Schedule
FC	fuel cell	WLTP	World Harmonized Light Vehicle Test Procedure
RO	Robust optimization	EES	Electrical energy storage system
EREV	Extender range electric vehicle	TES	Thermal energy storage
EF	Equivalent factor	QRDCNN	Quantile Regression Dilated Causal Convolution Neural Network
SOC	State of charge	QRNN	Quantile regression neural network
PID-ECMS	Proportional integral derivative ECMS	QEMAVOA	Quantum-inspired African vultures optimization algorithm with an elite mutation strategy
S-ECMS	Shooting method ECMS	QC	Quantum Computing
A-ECMS	Adaptive ECMS	QDCE	Quantum double-chain encoding
UC	Ultra capacitor	QRG	Quantum rotating gate
		OBL	Opposition Based Learning

into the DS framework to optimize DS and EVCS operations (Ebrahimi et al., 2020a). Intelligent dispatch techniques for EVCSs help mitigate their impact on DS load demands (Zengin et al., 2016a). Coordinated management strategies between the DS and EVCSs can achieve peak demand shaving, improve voltage profiles, and reduce power losses (Singh and Letha, 2018a). Moreover, in a DS environment, smart energy EVCSs aim to minimize operational costs (Savio et al., 2019a).

Home Energy Management System (HEMS) in (Shojil et al., 2014) prioritizes residential suitability by efficiently managing many home appliances, including EVs without considering grid impact and Renewable Energy Source (RES) penetration. The author of (Rastegar et al., 2016; Karfopoulos and Hatziaargyriou, 2012; Venkatesh and Guan L., 2012; Wen et al., 2012) proposes Energy Management System (EMS) that utilizes EVs to minimize residential operating costs. Each EV charging plan was upgraded to meet separate goals, an aggregator coordinates these plans to reduce impact of EVs charge on grid. These systems do not evaluate the potential use of RES. HEMS that integrate Photovoltaic (PV) systems and EVs are described, focusing on optimizing household energy use without assessing the impact on the grid. Therefore, the potential reduction in profits due to decreased PV energy availability during voltage restrictions is not considered (Erdinc et al., 2014; Ito et al., 2017; Kawashima et al., 2015). To reduce residential operating costs, EVs must be charged when Photo Voltaic generation was low and is executed to cover the residential electricity usage when Photo Voltaic generation was high. High Photo Voltaic energy was sold back to grid on the basis of Feed in Tariff Program.

Photo Voltaic reduction to mitigate overvoltage from surplus PV production poses a significant challenge (Gholinejad et al., 2020; Nazari-Heris et al., 2020). This reduction can lead to a loss in expected profits from selling surplus PV production. To effectively utilize PV output in Vehicle-to-Home (V2H) scenarios, a framework to manage EV

charging and discharging is required to reduce PV restrictions caused by grid voltage limitations (Ganjeh Ganjehlou et al., 2020). However, they have not considered the impact of uncertainty in predicted power profiles used for charging and discharging plans (Wu et al., 2016b). In recent years, a variety of energy management strategies has emerged, and is categorized into the approaches which is based on rule and optimization. Strategy based on rule was developed on the basis of power component characteristics along with specific driving conditions. Their simple control frameworks make them easy to apply in engineering practice. However, these strategies depend heavily on the logic of rule setting which require frequent calibrations. Control rule placed solely on human experiences may not extensively cover all driving condition, leading to potential inefficiencies. In contrast, optimization-based methods have gained traction in energy management due to their ability to optimize power allocation and minimize fuel intake across various driving condition (Thota et al., 2020b).

Optimization-based methods, like Dynamic Programming (DP) (Hannan et al., 2018), Pontryagin's Minimum Principle (PMP) (Ji et al., 2016b; Liu et al., 2019b), Model Predictive Controls (Abbasi et al., 2019b), and ECMS (Ebrahimi et al., 2020b; Zengin et al., 2016b) were commonly used for global and real-time development controls due to their strong optimization capabilities (Singh and Letha, 2018b). DP requires knowledge of the entire driving cycle for optimal control strategy, making it unsuitable for dynamically changing conditions (Savio et al., 2019b). As driving cycles often change in real-world scenarios, DP may not be able to solve accurate energy management system (EMS) due to the future drive cycles which is unknown. Pontryagin's Minimum Principle depends on Hamilton function, although its complex multi-lateral constraint can impede optimization efficiency, leading to its use as a benchmark for evaluating other algorithms. ECMS, on the other hand, is a cost-relied and an online optimization method known for its

real time control(RTC) capabilities which was successfully used to many commercial use vehicle, making this as a benchmark which is the reference for other real time control(RTC) method (Torreglosa et al., 2016; Yang et al., 2023a).

The objective of this research work is to develop a novel hybrid technique, Quantum VultureNet, for optimizing energy management systems (EMS) integrating EVs and distribution systems (DS). Quantum VultureNet combines the African Vultures Optimization algorithm with elite mutation, dynamic opposition-based learning, and chaotic map-based population initialization. This approach aims to minimize system costs and power losses while optimizing energy management strategies, particularly focusing on factors like bidirectional energy trading, EV fleet arrival times, uncertainties in Photovoltaic (PV) power generation, and influences affecting energy selling back to the grid. Through simulation studies, the effectiveness of Quantum VultureNet is demonstrated in finding near-global optimum solutions with reduced computational complexity compared to conventional methods.

Motivation:

EVs rapidly gained popularity recently, leading to an expansion of EV Charging Stations and creating finer platform for exchanging energy between EVs and power grids. However, Electric Vehicle Charging Stations (EVCS) presents significant challenge to power grid due to their unpredictable load characteristics, especially in non-residential areas, such as shopping mall, company site, or car rental station. Ensuring the safe and continuous working of the electric grids, Distribution System operator requires reliable load prediction for EVCS. The rapid growth of EVs increases power demand adding to utility grid loads and causing fluctuations that hinder EV penetration. Several techniques, like maximum-weight EVs dispatching algorithms, optimal price algorithms, and two-staged charge strategies were advanced to mitigate effect of EM between EVs and Distribution System. Additionally, riffled algorithms can establish upper limit profiles for EVCS energy intake, enabling grid stability in an efficient and cost-effective way. However, existing systems lack the ability to use additional storage units to compensate for EV uncertainties, motivating further research in this area.

The research's primary contribution is:

The significant contributions of the proposed work include:

- Develop a sophisticated Energy management setup for EV using Quantile Deep Learning with Enhanced African Vulture Optimization.
- Create the Quantum VultureNet strategy, a hybrid approach integrating Quantile Deep Learning and Enhanced African Vulture Optimization (EAVO), to identify optimal system parameters.
- Address the issue of random initialization in the algorithm by proposing Elite Mutation, Dynamic Opposition-Based Learning (DOL), and Chaotic mapping-based population initialization in EAVO Algorithm.

Organization of the paper:

This paper was arranged as given: Part 2 conducts recent work's thorough literature survey in field. Section 3 delves into system model and design analytics. Part 4 outlines devised method along with their implementation method. Part 5 presents result of simulation conducted using the proposed approach. At last, Part 6 wraps up the paper with a concluding area.

2. Literature review

2.1. Related work

In 2021, B. Rajani and D. Chandra Sekhar (Rajani and Sekhar, 2021) introduced the FFA-GBDT technique for EMS in EV and DS, aiming to minimize system costs and power losses while utilizing energy management (EM). The technique incorporates EMS operation, including energy trading in both directions, EVs fleet arrival times, PV uncertainty

impact, and prioritization factors, demonstrating effectiveness in achieving near-global optimum solutions with reduced complexity and energy consumption of 720.34 KJ.

In 2020, Rayhane Koubaa et al (Koubaa et al., 2020). developed the energy management system for FCHEV that considers uncertainties. They focused on fuel economy, the slow dynamics of Fuel Cell network, robustness of solutions as key performance criteria. The study addressed uncertainties related to power productions, conversions, and demand level, acknowledging various sources of errors in operation conditions, modelling, and estimation. The research investigated a Robust Optimization based Energy Management System to enable optimum and efficient performance despite uncertain parameter. To address concerns about RO's over-conservatism, the study introduced conservatism level parameters to allow for flexible decision-making and effective performance.

In 2020, Ye Yang et al (Yang et al., 2020). made the adaptive optimal real time EMS. Their approach included the improvised shooting algorithm to determine the Electric Fraction (EF) range based on vehicle power configuration parameters. They also employed secant method to calculate the initial EF value faster. Additionally, they analyzed operational working of ECMS control strategies from an energy flow perspective, elucidating the relationships among the five operating modes of EREVs.

In 2021, Mingyao Yao et al (Yao et al., 2021). applied quadratic program-based analytical solution method to crack the Electric Fraction and introduced adaptive update procedure for terminal SOC in correcting prediction error in driving cycle. They then presented and validated an A-ECMS for optimal control in real-time in Extended-Range EVs. Simulation result demonstrated that Adaptive-ECMS could be maintained in a terminal State of Charge close to targeted value, which is termed in fuel economies, variation between A-ECMS and a dynamic program-based global optimization Energy Management System was lower than 2%.

In 2021, Achikkulath Prasanthi et al (Prasanthi et al., 2021). improved the optimal sizing methodology in hybrid energy sourcing of HEVs that include Ultra capacitors (UC), Fuel Cells (FC), and Batteries (BU). The authors formulated multi-objective problems using dynamic-source model to assess initial cost of system, weight, running cost, also cost associated with source degradations. Additionally, they proposed a novel Adaptive Energy Management System which considers effective-sourced properties and also the driving power of cycle demands as a frame of improvisation problems. Solving hybrid energy source optimization problems, they enhanced Biogeography-Based Optimization Algorithm by incorporating quantum wave concepts to explore search space more efficiently.

In 2021, Zhuoran Hou et al (Hou et al., 2021). devised an innovative Energy Management System with enhanced adaptability to Vehicle (PHEVs). Control parameters, obtained from benchmark tests, was calculated offline for distinct driving condition. This optimized parameter is then applied based on real-time identification of different driving behaviours. This offline-online cooperation enhances EMS performance across a variety of driving conditions.

In 2021, Guodong Du et al (Du et al., 2021b). developed an experimental Deep Reinforcement Learning control strategies for energy management in SHEVs. They introduced the vehicle's powertrain model and EMS formulas, followed by constructing a comprehensive control structure with nest looping logic for Energy Management System. Within the structure, they proposed Hybrid Experience Replay to improve experienced sampling and efficiency of training. Additionally, they presented NAG-Adams optimization to enhance optimization effectiveness.

In 2020, Guodong Du and collaborators (Du et al., 2020) introduced two novel algorithms, one based on Reinforcement Learning (RL) and the other on Deep RL (DRL), for managing energy in a SHETV. They initially constructed powertrain model of SHETV therefore established corresponding energy management (EM) formula. They then developed

Table 1
Advantages and Gap Addressed in Literature review.

Study	Gap Addressed	Benefits	Utilization by DSOs
Rajani and Chandra Sekhar (Rajani and Sekhar, 2021)	Minimization of system costs and power losses in EMS for EV and DS	Reduced system costs and power losses, effective energy management	Improved efficiency and cost-effectiveness of EMS for EV and DS, potential benefits for grid stability and energy distribution
Koubaa et al (Koubaa et al., 2020).	Uncertainties in FCHEV energy management	Improved fuel economy, robustness in Fuel Cell network dynamics	Improved energy management in FCHEVs, potentially leading to longer vehicle range and increased reliability
Yang et al (Yang et al., 2020).	Adaptive optimal real-time EMS, EREV energy flow analysis	Improved determination of Electric Fraction (EF) range, faster calculation of initial EF value	Enhanced real-time control strategies for EREVs, potentially leading to increased efficiency and performance
Yao et al (Yao et al., 2021).	Analytical solution method for Electric Fraction, A-ECMS for ER-EVs	Improved prediction error correction in driving cycle, maintenance of terminal SOC close to targeted value	Improved optimal control in real-time for ER-EVs, potentially leading to improved fuel economies
Prasanthi et al (Prasanthi et al., 2021).	Optimal sizing methodology for hybrid energy sourcing in HEVs	Improved initial cost assessment, weight reduction, running cost optimization	Enhanced hybrid energy source optimization, potentially leading to cost-effective and efficient HEVs
Hou et al (Hou et al., 2021).	Adaptive Energy Management System for PHEVs	Enhanced adaptability to driving conditions, offline-online cooperation for EMS	Improved EMS performance across various driving conditions, potentially leading to increased efficiency and adaptability
Du et al (Du et al., 2021b).	Deep Reinforcement Learning control strategies for energy management in SHEVs	Improved EMS formulas, comprehensive control structure with nest looping logic	Enhanced control strategies for SHEVs, potentially leading to improved efficiency and performance
Du et al (Du et al., 2020).	RL and DRL algorithms for managing energy in SHETVs	Novel algorithms for energy management in SHETVs, efficiency improvements	Improved energy management in SHETVs, potentially leading to increased efficiency and performance
Zhou et al (Zhou et al., 2020).	EMS design for P-HEV with HESS	Efficient power flow management, utilization of Decentralized Fuzzy Logic Controller	Enhanced power flow management in P-HEVs, potentially leading to increased efficiency and performance
Mehrabi et al (Mehrabi et al., 2020).	Decentralized algorithm for directing EVs to Charging Stations	Enhanced EV charging efficiency, effective utilization of charging stations	Improved EV charging strategies, potentially leading to reduced charging times and improved grid management

new model of the Reinforcement Learning(RL) method named Dyna which combines empirical plan using Dyna agent and thus it was applied to EM controls for the SHETV. Efficiency using Dyna-H was confirmed by comparing it with the Dynamic Programming and the conventional Dyna methods.

In 2020, Shiyao Zhou et al (Zhou et al., 2020). developed the EMS designs and efficient method for P-HEV with HESS comprising a Lithium-Titanium-Oxygen battery package and Nickel-Cobalt-Manganese battery packs. EMS manages power flow within hybrid powertrain which utilizes Decentralized Fuzzy Logic Controller that takes prediction for PHEV powertrain state as inputs.

In 2020, Abbas Mehrabi et al (Mehrabi et al., 2020). devised a low-complexity online decentralized algorithm that employs effective heuristics to efficiently direct each EVsto the most profitable Charging Station (CS). Simulation results on IEEE 37 bus distribution networks demonstrate that stated algorithms enhances social aid by approximately 30 % average compared to alternative scheduling strategies when EVs participate equally in charging and discharging operations.

2.2. Problem statement

Several studies have explored Energy Management Systems (EMS) for EVs and DS, engaging several techniques and aspect is shown in Table 1. The hybrid FFA-GBDT system aims to reduce system costs and power losses while optimizing energy management. However, in multimodal optimization, the Firefly Algorithm (FFA) may struggle to efficiently locate all relevant optima. Handling high-dimensional optimization problems remain challenging for many nature-inspired algorithms, including FFA. Moreover, training a Gradient Boosting Decision Tree (GBDT) model can be computationally expensive. The adopted Robust Optimization based algorithms considers uncertainty in cost functions and constraint, safeguarding system performances from possibility and optimality issue. The Adaptive ECMSshows better fuel economy than PID-ECMSand better real-time performance than S-ECMS. However, the algorithm's efficiency may decrease with increasing dimensionality. Using Deep Q-Learning (DQL) with the AMS Grad optimizer can lead to better optimization effects, although with additional computational costs compared to basic gradient descent. Implementing fitness functions for multi-objective optimizations in Energy Management System under the Worldwide Light-duty Test Cycle requires incorporating decision-makers' preferences. The choice of weights, priorities, and preferences can significantly affect the final solution set. The online decentralized algorithms with the low complexity efficiently direct each EVs to most profitable Charging Station. However, decentralized systems may lack robust data backup mechanisms, making recovery challenging in case of node failure or data loss. Table 1 shows significant research gaps remain in the area of the EM between EVs EDS that needs to be addressed. The major considerations include,

- Research on EM between EV along with the EDS to attain objective lacks.
- Research on investigating optimal energy management with cost consideration is scarce.
- There has not been much focus on computation and the complexity reduction of the system model.
- Inadequate attention is known about how optimization strategies are always necessary for energy management.

To address the limitations in existing literature, this study proposes the Quantum VultureNet model, which integrates Quantile Deep Learning and Enhanced African Vulture Optimization (EAVO). The EAVO algorithm combines Quantum Inspired African Vulture Optimization (AVO) technique along with Elite Mutation Strategy, dynamic opposition-based learning, and chaotic mapping-based population initialization. EAVO is a metaheuristic optimization algorithm. The concept of quantiles is incorporated into the deep learning model to

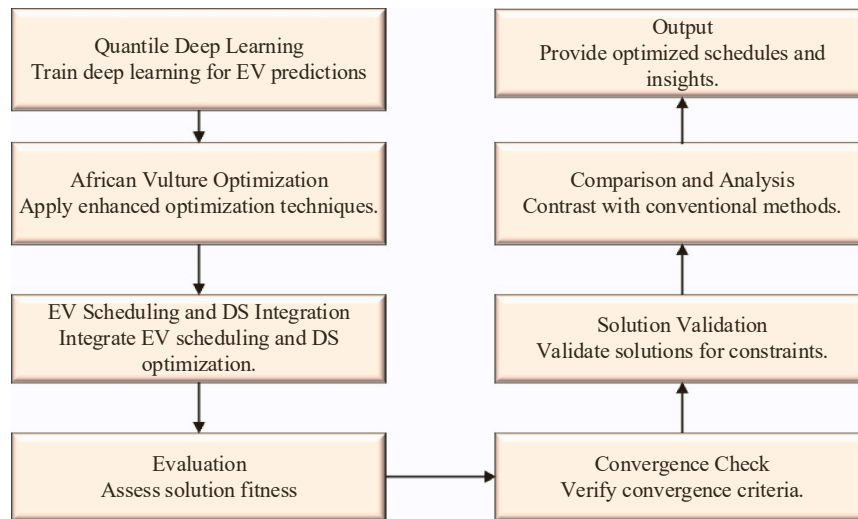


Fig. 1. Flowchart of the Research Stages.

construct the EMS quantile prediction component.

3. System modelling and design analysis

The EM system which was introduced for EVs and DS in the Fig. 1 includes PV panels, batteries, an ESS, converters, loads, the utility grid, and EVs. Mathematical models for PV panels, batteries, loads, ESS, and EVs are explained, with the EV model consisting of a driver model, powertrain model, and vehicle dynamics model. The DC-AC converter in the EMS converts DC from the battery or solar panels to AC for the load. The Quantum VultureNet framework includes a Training Phase using quantile deep learning model and a Testing Phase using EAVO for cost-effective predictions. This method optimizes control variables for the EMS. African Vultures Optimization (AVOA) is chosen for its effectiveness in solving complex problems, particularly for its ability to find global optima in scenarios with numerous variables and challenges, mirroring the efficient foraging behavior of African vultures. Its proven performance, demonstrated through research, showcases its superiority over other optimization algorithms across various benchmark functions and real-world engineering problems. AVOA's versatility allows it to be applied in diverse fields such as machine learning, power systems, and resource allocation. Its nature-inspired approach, based on vultures' natural behavior, makes it intuitive to understand and implement. Additionally, ongoing research has led to the development of enhanced versions of AVOA, like EAVOA, which incorporate additional strategies for improved performance, contributing to its status as an active and evolving research area.

3.1. EV modelling

Simulation plays a major role in energy management of EV systems, with an accurate EV model being the foundation for such simulations shown in Fig. 2. The procedure for vehicle modeling is detailed in following subsection.

3.1.1. The driver models

Driver models take current speed of vehicle as an input and determines the brakes and accelerator pedal's positions based on negative or positive speed errors. A Proportional-Integral feedback control was employed for the pedals in achieving desired braking and accelerations, which is expressed as follows:

$$d_r = Ae_B(h) + A_a \int_0^h e_B(\tau) d\tau \quad (1)$$

The outcomes of PI control were constrained within ranges between -1 and 1 . When value of $dis(0, 1]$, position of the accelerator pedal is set to $[0\%, 100\%]$ as below:

$$\theta_s = \begin{cases} d_r, & \text{if } d_r > 0 \\ 0, & \text{if } d_r \leq 0 \end{cases} \quad (2)$$

When d_r 's value is $(-1, 0]$, the pedal position of brake was fixed as $[100, 0]$ as given below:

$$\theta_t = \begin{cases} -d_r, & \text{if } d_r > 0 \\ 0, & \text{if } d_r \leq 0 \end{cases} \quad (3)$$

When value of d becomes 0 , both acceleration and braking pedal is set to 0 . The gain is determined based on UDDS and WLTP driving cycle.

3.1.2. Engine models

Combustion engine internally provides power to drive vehicles or battery charging. However, when an engine runs at inert position, thereby consumes fuels without producing the power, making idle condition zero efficiency point. The consumption of fuel rates for the engines were given as follows:

$$P_H = e_\eta(\omega_x, S_x) \quad (4)$$

where, P_H implies consumption of fuel masses through engine, ω_x denotes the speed of engines, S_x denotes the torque of engine, and also e_η denote monotone functions.

3.1.3. Battery model

Internal resistance N_a is known by State of Charge functions. Battery outcome of V-I is calculated by Eqs. (5), (6).

$$d_r = d_{is} - y_t N_a \quad (5)$$

$$y_t = \frac{m_r}{d_r} \quad (6)$$

where, y_r denotes the battery current, d_r denotes the ultimate battery voltages, m_r denotes the output battery power, N_a denotes internal resistance of the battery, d_{is} implies open circuit voltage. The open circuit voltage d_{is} denotes temperature function and the State of Charge and was denoted by Eq. (7).

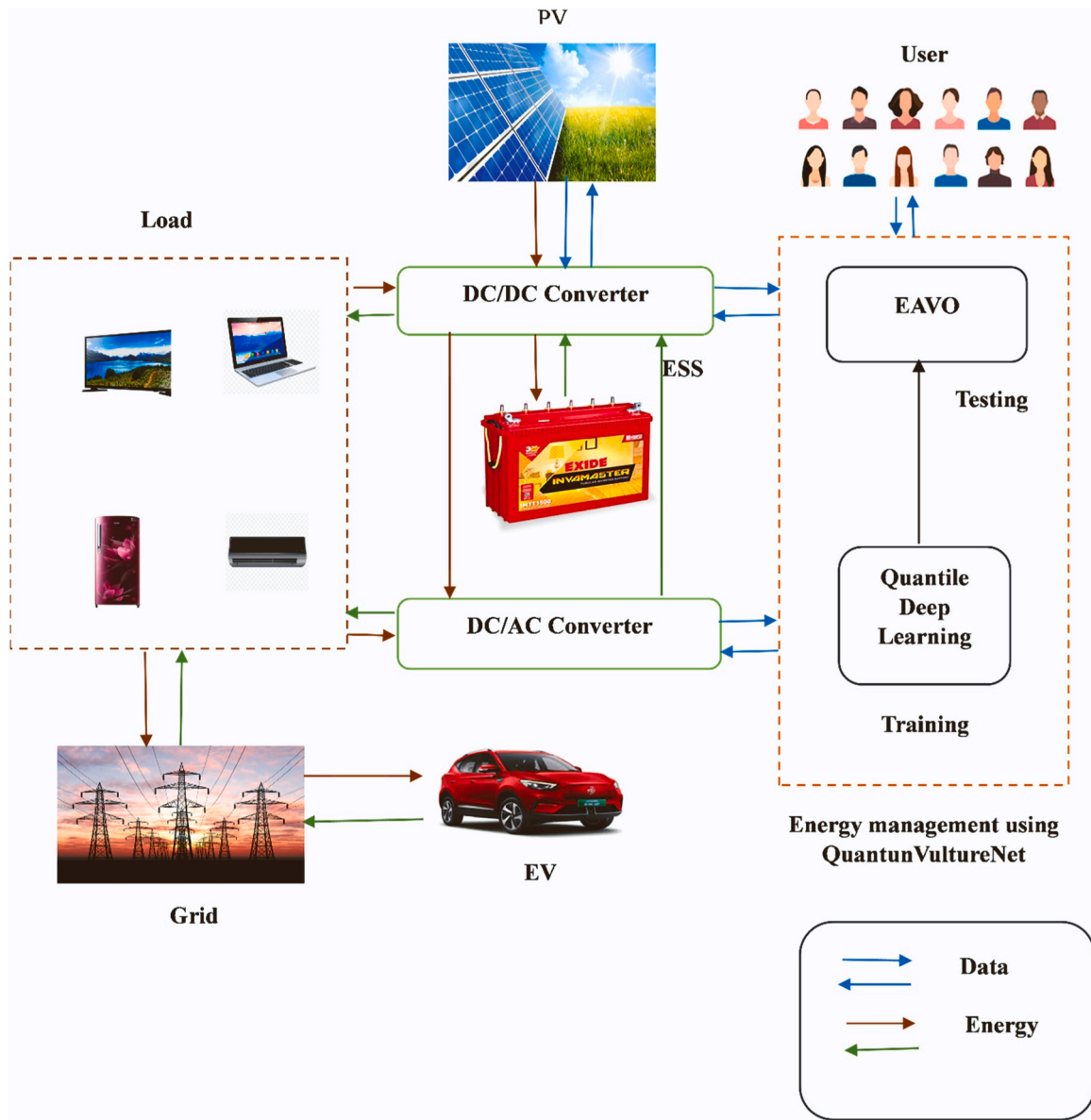


Fig. 2. Schematic representation of the introduced EM in EVs.

$$d_{ts} = \alpha_0 + \alpha_1 soc + \alpha_2 soc^2 + \alpha_3 S^* + \alpha_4 soc S^* + \alpha_5 soc^2 S^* \quad (7)$$

where, $(\alpha_0, \dots, \alpha_5)$ implies the coefficient needs to be determined using experimental data. Internal resistance of battery, represented by N_a , is the function of temperature and State of Charge, and can be given as:

$$N_a = \beta_0 + \beta_1 soc + \beta_2 soc^2 + \beta_3 S^* + \beta_4 soc S^* + \beta_5 soc^2 S^* \quad (8)$$

where, $(\beta_0, \dots, \beta_5)$ denotes the coefficients that are identified through experimental data. Assessing SOC of battery is vital for the accuracy of battery's model. Errors in SOC estimation can arise from incorrect initial SOC estimation, current measurement errors, and accumulated errors. Therefore, constructing an accurate SOC calculation models was essential. Here, Coulomb counting technique is employed in evaluating SOC because of the popularity and higher precision. Battery SOC was evaluated on the basis using the Eq. (9).

$$soc(h) = soc(o) - \int_0^h \frac{y_1(\tau) d\tau}{m_{u_i}} \quad (9)$$

where, $soc(0)$ refers SOC initial values and P_m refers to capacity of nominal batteries. Here lower and the upper limits of battery SOC was made as 65 % along with 95 % in reducing model degradation effect at lower State of Charge along with aging of battery at higher State of Charge.

3.2. Vehicle dynamic model

The dynamic model of the vehicle was primarily accountable for adjusting torque from engine and electric motors to generate the pulling forces needed for wheel movement. Subsequently, aerodynamic forces, rolling resistances, and the road grades are joined to calibrate vehicle speed. This force balance is represented by the Eq. (10).

$$b_{gs} = b_g + b_i + b_j + b_h \quad (10)$$

where, b_{gs} denotes the acceleration force of vehicles, b_h implies traction force of vehicles, b_i refer the rolling force resistances, b_j imply of road's grade force, b_g is aerodynamic resistance force. The rolling resistance force along with the resistance of aerodynamic were the major dragging force, while force of traction denotes the drive force given through

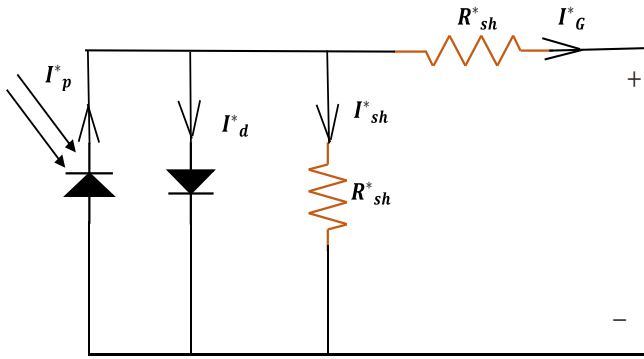


Fig. 3. PV system model.

vehicle's power trains. The road's grade forces are the dragging force based on the road slope. The vehicle's acceleration force b_{gs} was illustrated using Eq. (11).

$$b_{gs} = Dg \quad (11)$$

$$h = B_{ksp} \quad (12)$$

The force of aerodynamic was denoted by equation given below.

$$b_g = \frac{1}{2} \mu G_{gl} F^* B_{ksp} \quad (13)$$

where, μ is density of air, G_{gl} is air drag coefficient, F^* represents vehicle front area F , B_{ksp} shows vehicles velocity. Other than this, rolling resistance force was represented by Eq. (14).

$$b_i = \cos(\beta) K_i D p^* \quad (14)$$

where, β refers to road slope function, K_i denotes rolling resistances, D denotes weight of vehicles, and p^* denotes the gravity constant. The road's grade force was calibrated by the Eq. (15).

$$b_j = \sin(\beta) D p^* \quad (15)$$

The traction force of vehicle b_h was denoted by Eq. (16).

$$b_h = b_{IFK} + b_{KD} \quad (16)$$

where, b_{KD} implies the electric generator/motor force and b_{IsK} implies force of engine. The engine force b_{IFK} was eventually dealt with the speed and torque of engine. The force of engine which is constructed as the wheel's torque and radius is denoted by below equation.

$$b_{IFK} = \frac{S_X}{N_X} \quad (17)$$

where, N_X is radius of wheel and S_X wheel's torque and is estimated using Eq. (18).

$$S_X = S_{higv} N_{hvj} \zeta \rho_{hvm} \quad (18)$$

where, S_{higv} is torque input transmission, N_{hvj} denotes the transmission gear ratio, ζ indicate ratio of ending drive gear, ρ_{hvm} shows transmission path efficiency. The of engine's outcome torque was taken identical as transmission torque input as shown:

$$S_{higv} = S_{IFK} \quad (19)$$

where, S_{IFK} is torque of the engine. Fuel usage of the engine is interpreted as map fuel as the engine speed and torque's purpose is given:

$$P_{IFK} = b(\omega_{IFK}, S_{IFK}) \quad (20)$$

where, ω_{IFK} is engine speed. The engine speed was represented in following equation.

$$\omega_{IFK} = \omega_{higv} \quad (21)$$

$$\omega_{higv} = \omega_X N_{hvj} \zeta \quad (22)$$

$$\omega_X = \frac{B_{ksp}}{N_w} \quad (23)$$

The Energy Management force was calculated from wheel's torque as shown:

$$H_{EM} = \frac{S_X}{N_X} \quad (24)$$

$$S_X = S_{EM} \rho_{EM} N_{EM} \quad (25)$$

The battery's outcome power of was basis of discharging and charging condition. The outcome power of battery depends on the Energy Management torque functions, Energy Management efficiency, and Energy Management speed, which was described as following Eq. (26).

$$m_l = \begin{cases} \frac{\omega_{EM} S_{EM}}{\rho_{EM}}, & \text{discharge} \\ \omega_{EM} \rho_{EM} S_{EM}, & \text{charge} \end{cases} \quad (26)$$

The Energy Management torque, efficiency, and speed was considered based on equations below:

$$\rho_{EM} = b(\omega_{EM}, S_{EM}) \quad (27)$$

$$\omega_{EM} = \omega_{X,i} N_{EM} \quad (28)$$

$$\omega_{X,i} = \frac{B_{ksp}}{N_X} \quad (29)$$

The vehicle's entire torque demand was calculated using driver command, which was acceleration pedal location, and is mentioned by Eq. (30).

$$S_r = \theta_s S_i \quad (30)$$

The whole torque demand is divided through the Energy Management System which is illustrated by Eq. (31).

$$S_r = S_{r,IFK} + S_{r,EM} \quad (31)$$

The totality EM torque comprises demand of torque given through the EMS and the brake torque demands.

$$S_l = \theta_l S_{EM, \min} \quad (32)$$

$$S_{r,EM,h} = S_{r,EM} + S_l \quad (33)$$

where, S_l is torque's brake demand.

3.3. The modelling of Photo Voltaic systems

Solar cell is the fundamental component of a Photo Voltaic unit used to generate electricity. PV cell is connected in either parallel or series pattern forming solar panels with necessary specifications. Fig. 3 illustrates circuit of PV systems. Here in the PV energy method, Photo Voltaic panel converts incoming sunlight to electrical energy. Power produced by Photo Voltaic method was calculated using the Eq. (34).

$$P_p = N_{hp} \rho_p G_p \quad (34)$$

where, P_p represents Photo Voltaic panel output power, N_{hp} represents solar radiation in the tilted plane modules, ρ_p indicates PV panelefficiency, G_p is the Photo Voltaic panel area. The equation below articulates Photo Voltaic panel efficiency,

$$\rho_p = \rho_{iu-p} \rho_{YF} [1 - D_s(S_{cell} - S_i)] \quad (35)$$

where, efficiency of Photo Voltaic panel represents ρ_p , efficiency of reference module represents ρ_{ref} , efficiency of power condition represents ρ_{PF} , temperature coefficient of PV panel efficiency was D_s , the temperature of cell $^{\circ}C$ is S_{cell} , and reference condition cell temperature was S_r . Suppose an ideal power tracker was utilized, ρ_{PF} values were equal to 1. Likewise, S_r was placed at $25^{\circ}C$ and mono and polycrystalline silicon the temperature D_s as $-3.7 \times 10^{-3}^{\circ}C^{-1}$. Also, S_{cell} articulated as below:

$$S_{cell} = S_{gu} + \left[\frac{S_{PF} - 20}{800} \right] N_{hp} \quad (36)$$

where S_{gu} represents ambient air temperature, S_{PF} represents operating temperature of the nominal cell.

The Photo Voltaic Module V-I relation was denoted by Eq. (37).

$$R^* = R^*_J - R^*_{fg} \left\{ \exp \left[p^* \frac{Q^*_p + R^*_p \cdot N^*_{fb}}{GUS_{cell}} - 1 \right] \right\} - \frac{Q^*_p + R^*_p \cdot N^*_{fb}}{N^*_{fp}} \quad (37)$$

Where, R^*_J represents light of current generated in G, N^*_{fb} along with N^*_{fp} are series and shunt resistance in Ω , G shows ideal factors, U denotes Boltzman constant, R^*_p and Q^*_p represents solar cell output current, voltage and R^*_{fg} shows reverse saturation current in G.

Current due to Light-generation R^*_J and the reverse saturation current R^*_{fg} were given in Eqs. (38) and (39).

$$R^*_J = [R_{fs} + \alpha(S_G - S^*_G)] \frac{G_S}{G^*_S} \quad (38)$$

$$R^*_{fg} = \frac{R_{fs} \alpha (S_G - S^*_G)}{\exp \left[\frac{Q_{fs} \alpha (S_G - S^*_G)}{Q} \right] - 1} \quad (39)$$

where, S_G shows of Photo Voltaic cell ambient temperature, G_S denotes the irradiance, α denotes the temperature coefficient of short circuited current, Q_{fs} denotes open-circuited voltage at specific temperatures, Q_{Sp} indicate voltage due to thermal power. Here, a preliminary series of the six cluster centres have been chosen, i.e., $\sigma = \beta_{h1}, \beta_{h2}, \dots, \beta_{hi} h \in S$. Here, σ and β represents the number of circumstances for generation of Photo Voltaic and Photovoltaic generation indices.

In our proposed Quantum VultureNet technique, we model and integrate PV uncertainties by considering historical data and weather forecasts. We employ a probabilistic approach to simulate various possible scenarios of solar power generation based on these uncertainties. This allows us to create a range of possible outcomes for solar power availability, which is crucial for accurate energy management. The scheduling and optimization process in Quantum VultureNet then takes these uncertainties into account, optimizing the energy management system to adapt to different levels of solar power generation. By incorporating PV uncertainties in this manner, our approach enhances the robustness and adaptability of the system to dynamic environmental conditions, ultimately improving its performance in managing EV energy needs within the distribution system.

3.4. Modelling of electrical load

Based on feature, load was classified into non-dispatchable load, shiftable load, and interruptible load (Adouni et al., 2016; Ali et al., 2018). The load general constraint which was shiftable is expressed by Eq. (40).

$$0 \leq E^i_z \leq E^{i-}_z \quad \forall z$$

$$\sum_{z \in \mathcal{N}} E^i_z = E^{i,X} \quad (40)$$

Where, $E^{i,X}$ is fully shifting load in total period during $z \in E$. The entire cost

related with interrupted load is expressed using Eq. (41).

$$SF_{yv} = \sum_{z \in \mathcal{N}} \mu^y_z E^y_z \Delta U^* \quad (41)$$

The interrupted load upper bound is represented by Eq. (42).

$$0 \leq E^y_z \leq E^{y-}_z \quad \forall z \quad (42)$$

For preventing entire suitable load, load curves were in limit and was transformed into identical periods, as in Eqs. (43) and (44).

$$W^{To}_k E^y_z = W^m_k + W^{sh}_k + W^{in}_k - W^{in}_k \quad \forall k \quad (43)$$

$$(1 - \beta_2) W^{To}_k \leq W^{To}_k \leq (1 + \beta_1) W^{To}_k \quad \forall k \quad (44)$$

where, W^m_k is the non-dispatchable load in period k , W^{To}_k is distinctive electricity load previous than the demands in kW , β_1 and β_2 were two positive parameter denoting response space's electricity demand. Using electrical load model, variables set is denoted using:

$$X_{EL} = \{TC_{in}, W^{in}_k, W^{To}_k, W^{sh}_k, W^m_k\} \quad \forall k \quad (45)$$

3.5. Modelling of ESS

This study considers ESS on basis of both chemistry and TES. In the TES system, storage of heat is done by utilizing water as the heat storage medium. Operation and service cost of Thermal Energy Storage model is minimized. Degradation of ESS on the basis of chemistry is designed using linear functions, as denoted using Eq. (46).

$$IF_{ES} = \sum_{i \in \mathcal{I}} H_{ES} \left(P^A_i + P^B_i \right) \Delta I^* \quad (46)$$

Where, H_{ES} is the degradation of linear EES cost co-efficient ($\$/kWh$), P^A_i and P^B_i implies charging and discharging power in period $i(kW)$. Here, in the Thermal Energy Storage, energy storage is done in the form as water is utilized from heat storing medium of Thermal Energy Storage. Thus, Thermal Energy Storage preservation and operational cost were neglected. Stored energy of Energy Storage System and Thermal Energy Storage system was represented by following eqns.

$$E^A_i = E^A_{i-1} + (\eta_A P^A_i - P^B_i / \eta_B) \Delta I^* \quad \forall i \quad (47)$$

$$H^I_i = (1 - \lambda \Delta I^*) H^I_{i-1} + (\eta_{HA} H^A_i - H^B_i / \eta_{HB}) \Delta I^* \quad \forall i \quad (48)$$

Where, η_A / η_B and η_{HA} / η_{HB} are charging/discharging efficiency EES and Thermal Energy Storage. The limit of energy and charging/ discharging of EES and Thermal Energy Storage system are explained in Eqs. (49) and (50).

$$0 \leq P^c_i \leq P^{c-}_i, 0 \leq P^{ds}_i \leq P^{ds-}_i, E^A_i \leq E^A_i \leq E^{A-}_i \quad \forall i \quad (49)$$

$$0 \leq H^I_i \leq H^{I-}_i, 0 \leq H^{ds}_i \leq H^{ds-}_i, H^I_i \leq H^I_i \leq H^{I-}_i \quad \forall i \quad (50)$$

The charging/discharging efficiency may lead to increased losses that are non cost-effective. Therefore, ESS don't undergo charge and discharge. As a result, it was not required to use binary variable to estimate state of charge or discharge. Hence stored energy at end of development period was equal to the initial value, as demonstrated in Eq. (51).

$$E^A_{(\mathcal{N})} = E^A_0, H^I_{(\mathcal{N})} = H^I_0 \quad (51)$$

In Energy Storage model, variable set can be represented as:

$$X_{ES} = \{IF_{ES}, P^A_i, P^B_i, E^A_i, H^I_i, H^A_i, H^B_i, \forall i\} \quad (52)$$

3.6. Objective function

EVs attained total energy consumption at minimum which was denoted by Eq. (53) (Geng et al., 2020; Thomas et al., 2018; Thomas and Kotsakis, 2020).

$$I = \sum_{i=1}^n M(y(i), u(i)) \quad (53)$$

where $M(y(i), u(i))$ is mentioned as the cost function which was represented by following Eq. (54).

$$M(y(i), x(i)) = \sum_i \sum_{\omega} \Pi_{i\omega} \left(\frac{P_{i\omega}^{h,s}}{\Delta I} \cdot \varepsilon_i^d - \frac{P_{i\omega}^{h,j}}{\Delta I} \cdot \varepsilon_i^r \right) + \sum_i \sum_{\omega} \Pi_{i\omega} \left(\frac{P_{i\omega}^{p,j}}{\Delta I} \cdot \lambda_p + \sum_i \frac{P_{i\omega}^{E,j}}{\Delta I} \cdot \lambda_E + \frac{P_{i\omega}^{ES,j}}{\Delta I} \cdot \sum_i \frac{P_{i\omega}^{E,j}}{\Delta I} \cdot \lambda_{ES} \right) \quad (54)$$

Here, λ_p denotes Photo Voltaic priority parameter, λ_E represents EV priority parameter, λ_{ES} is Energy Storage System priority parameters, ε_i^d is daily electricity price in time i (cents/kWh), ε_i^r is daily price of electricity which is used for energy for selling grid in times i (cents/kWh), ΔI shows series of time interval during the period of 1 hour, ω denote Photo Voltaic generation scenario index, $P_{i\omega}^{h,j}$ is entire power given to the grid in kW, $P_{i\omega}^{h,s}$ is power requested as grid in kW, $P_{i\omega}^{p,j}$ is the power as Photovoltaic inserted in direction of grid in kW, $P_{i\omega}^{E,j}$ is the EV power inserted inside the grid in kW, and $P_{i\omega}^{ES,j}$ is the ESS power inserted to grid in kW.

3.7. Modelling of smart grid

Accurate modelling of Smart Grid elements is crucial for performance assessment and optimizing Energy-related systems. One key aspect is modelling the expected energy management, representing a Grid Scalability of the devices. This is often done using the Log-Normal distribution, a statistical tool for variables with logarithms following a normal distribution. Transforming variables through natural logarithms yields a log-normal distribution. The distribution PDF (Probability Density Function), defined by parameters μ and σ , shapes its characteristics. Leveraging the distribution captures the inherent energy management variability among Smart grid performances, benefiting applications like energy consumption forecasting, charging infrastructure planning, and Energy Management. The modelling strategy forms a solid foundation for comprehending and managing the stochastic nature of electricity usage patterns, enhancing EV management and deployment strategies.

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma y} \exp\left(-\frac{[\ln(y) - \mu]^2}{2\sigma^2}\right), y > 0 \quad (i)$$

where, μ and σ represents the parameters of location and scale, respectively, $f(x)$ represents PDF of a random variable y , y represents the random variable, $\exp$ represents exponential function, π represents the mathematical constant, $\ln$ represents the natural logarithm function. These parameters are computed using the following method:

$$\mu = \ln\left(\frac{m}{\sqrt{1 + \frac{v}{m^2}}}\right) \quad (ii)$$

$$\sigma = \sqrt{\ln\left(1 + \frac{v}{m^2}\right)} \quad (iii)$$

where, v and m represents derived mean and standard deviation. For modelling the probable daily mileage of EV, utilize a random variable y generated using the Bach-Muller method, this approach enables to

simulate and predict energy management strategy for Grids, taking into account the statistical characteristics of the data.

$$y = \sqrt{-2 \ln c_1} \times \cos(2\pi c_2) \quad (iv)$$

where, c_1 and c_2 are random variables, $\cos$ represents the cosine function. From the statistical data the electricity usage is calculated by following equation:

$$M_d = e^{(\mu_m + \sigma_m X)} \quad (v)$$

where, μ_m and σ_m represents the parameters of a PDF, M_d represents the expected electricity usage at grids. These parameters, μ_m and σ_m are determined based on statistical data. This statistical approach allows us to characterize and model the expected daily usage patterns of electricity in smart homes, facilitating predictions and analysis in the context of grid performance and usage.

Transitioning to a grid entirely composed of cables could bring about significant changes with both benefits and challenges. On the positive side, underground cables offer reduced environmental impact and enhanced aesthetic appeal by eliminating the need for towering structures and minimizing disruption to landscapes. They also promise greater reliability, as underground cables are less susceptible to weather-related outages and environmental hazards, thus bolstering the grid's resilience. Additionally, underground cables produce lower levels of electromagnetic interference, reducing potential disruptions to nearby electronic equipment. However, this transition poses notable challenges, including high upfront costs, technical complexities in installation and maintenance, and limited flexibility for future expansion and modifications. These challenges necessitate careful planning and investment to ensure the successful integration of underground cables into the grid infrastructure while maximizing their benefits and mitigating potential drawbacks.

4. Devised quantum vulturenet model for EM

The study proposes hybrid system for efficient EM between the EV and DS. The method combines Quantile Deep Learning and Enhanced African Vulture Optimization (EAVO), creating the Quantum VultureNet Algorithm. This hybrid algorithm integrates the enhanced AVOA with elite mutation, dynamic opposition-based learning (DOL), and chaotic mapping-based population initialization. The incorporation of quantile into the deep learning model enhances the algorithm's exploration and exploitation capabilities, particularly in dynamic environments, making it more effective for production scheduling problems. Quantum VultureNet, a novel concept in electrical power engineering, shows promise in various applications. It could optimize power flow, maintenance scheduling, and distributed energy resource management, leveraging its ability to find global optima. In smart grids, it could enable real-time data analysis for efficient operations. Additionally, its pattern recognition capabilities could enhance fault detection in power systems. The model offers global optimization similar to AVOA, scalability for large-scale tasks, and bio-inspired efficiency, potentially reducing computational costs. However, its practical implementation faces challenges due to hardware and software limitations, as quantum computing technology is still evolving, and integrating quantum algorithms with power systems requires expertise in both fields.

4.1. Step by step process for quantum vulturenet algorithm for EMS

The initial population of vultures was made uniformly, followed by random generation of EV parameter and calculation of fit functions. The best part was upgraded using Elite Mutation Strategy, dynamic opposition-based learning, and chaotic mapping-based population initialization. The cost function of EV is evaluated, thereby new vulture method was generated. Optimal values of the EV parameters, including

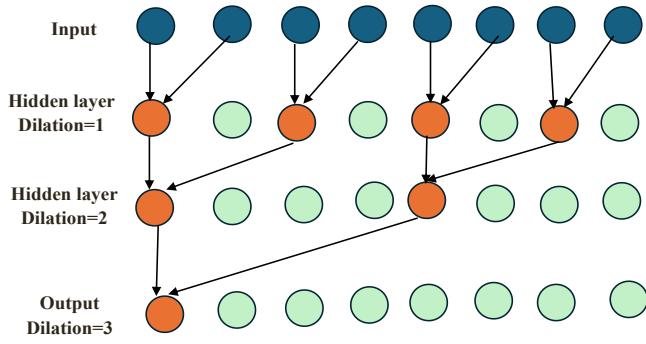


Fig. 4. One-dimensional dilation CNN.

battery state of charge, torque and the speed of engine, torque and speed of electromagnetics, and the power of battery, were determined by the proposed method for managing energy in the EVs system. This process was repeated iteratively until optimal solution is obtained.

4.1.1. Quantile deep learning

QRDCNN, a potent hybrid model, merges quantile regression with dilated causal convolutions. This model has proven effective in forecasting the probability density of EVsload.

4.1.1.1. Dilated causal convolution. One-dimensional EV power y_l is considered, based on previous EV series condition, the model with parameter θ is used in predicting future value $y(l+1)$. Thus an idea of causal system is made, where the system output is related to before value, rather than upcoming value. Output is denoted by Eq. (55), where M represent size of sample. Here, a dilated CNN is used to build the causal system (Wang et al., 2020; Xu et al., 2020).

$$q\left(y|\theta\right)=\prod_{l=0}^{M-1}q(y(l+1)|y(0),\dots,y(l),\theta) \quad (55)$$

To predict future EV power $y(l+1)$ through previous EV power series $y(0), \dots, y(l)$ is used as the input value and $x(t+1)$ as output value for model train, which is used in predicting Electric Vehicle power. An EVs power display a long-term performance. Thus, to allow networks to learn long-term dependencies, the structure of stack void convolution layer was adopted as in Fig. 4.

Here, intensive functions of predictive models are given in Eq. (56), where O and d are weights and bias terms of CNN.

$$F(o)=\frac{1}{2M}\sum_{l=0}^{M-1}(x_p-x_T)^2+\frac{\gamma}{2}(O,d)^2 \quad (56)$$

4.1.1.2. Quantile regressions. Considering sample $a=[a_1, a_2, \dots, a_m], b=[b_1, b_2, \dots, b_m]$, where a is response variable, and b is explanatory variable. Model parameter of linear regression was attained using objective function using Eq. (57), where β denotes regression coefficient (57).

$$\min \sum_{j=1}^m (x_j - y_j \beta) \quad (57)$$

Estimated problem of parameter vectors $\beta(\tau)$ using Eq. (57) is transferred into following optimization problem (Yang et al., 2023b).

$$\min_{\beta} \sum_{j=1}^M \rho_{\tau}(X_j - Y_j \beta) = \min_{\beta} \sum_{j|X_j \geq Y_j \beta} \tau |X_j - Y_j \beta| + \sum_{j|X_j < Y_j \beta} (1-\tau) |X_j - Y_j \beta| \quad (58)$$

Where, M is size of sample, X_j is true EV power, Y_j is input value of wind powers, $\rho_{\tau}(v) = v[\tau - J(v < 0)]$ represent loss function, and $J(\cdot)$ is indicative functions, as shown in Eq. (59).

$$J(v < 0) = \begin{cases} 1 & v < 0 \\ 0 & v \geq 0 \end{cases} \quad (59)$$

According to Eq. (58), several parametric estimation $\beta(\tau)$ is obtained under a several quantile τ which elaborates effects of variable on the quantile of responsible variable. When $\tau \in (0, 1)$ is analysed continuously, the distribution of the response variable is obtained, following conditional density. In the linear regression model in Eq. (58), relationship between the explanatory and response variable is linear. In many practical systems, the relation between variables were nonlinear (Hai et al., 2023).

$$P_X(\tau | Y) = e(Y, O(\tau), U(\tau)) \quad (60)$$

where $\dot{O}(\tau) = \{o_{ji}(\tau)\}_{j=1,2,\dots,I;i=1,2,\dots,I}$ represent the connected weight between the input and hidden layer, and $U(\tau) = \{u_{iz}(\tau)\}_{i=1,2,\dots,I;z=1,2,\dots,Z}$ represent the connected weight within layer. The QNN regression refer reducing Eq. (60) transfers the estimated parameter $O(\tau), U(\tau)$ into the optimization problem of Eq. (61).

$$\begin{aligned} \min_{O,U} & \left\{ \sum_{j=1}^M \rho_{\tau}[X_j - e(Y_j, O, U)] + \lambda_1 \sum_{j,i} o_{ji}^2 + \lambda_2 \sum_j u_j^2 \right\} \\ & = \min_{O,U} \left(\sum_{j|X_j \geq e(Y_j, O, U)} \tau |X_j - e(Y_j, O, U)| + \sum_{j|X_j < e(Y_j, O, U)} (1-\tau) |X_j - e(Y_j, O, U)| \right) \\ & \quad + \lambda_1 \sum_{j,i} o_{ji}^2 + \lambda_2 \sum_j u_j^2 \end{aligned} \quad (61)$$

where λ_1, λ_2 are penalty parameters. To avoid over-fitting process of model during trained phase, the Adam descent method was used in resolving Eq. (61) and thereby estimate the parameters vector $O(\tau), U(\tau)$.

4.1.1.3. Quantile Regressions Dilated Causal Convolutions. This study establishes a quantile regressive method of cavity-causality CNN based on the structure of the neural network regressive method and the quantile regressive method shown in Eq. (61). Cost function is transferred to objective function of quantile regression (Alsharif et al., 2023; Bianco et al., 2023).

$$\begin{aligned} e_{cost} &= \sum_{j=1}^M \rho_{\tau}[X_j - e(Y_j, O, d)] \\ &= \sum_{j|X_j \geq e(Y_j, O, d)} \tau |X_j - e(Y_j, O, d)| + \sum_{j|X_j < e(Y_j, O, d)} (1-\tau) |X_j - e(Y_j, O, d)| \end{aligned} \quad (62)$$

Finally, the parameter estimation is regarded as the optimization problem of Eq. (63).

$$\min_{O,d} e_{cost} + \frac{\lambda}{2} |O, d|^2 \quad (63)$$

Here, O and d denotes weight and bias set of a cavitation CNN. Also, the Adam descent method is used in resolving accuracy problems.

Next to solving parameters $\hat{O}(\tau), \hat{d}(\tau)$, we have to substitute them to Eq. (64) for obtaining conditional quantile estimations.

$$\hat{P}(\tau|Y) = e(Y, \hat{O}(\tau), \hat{d}(\tau)) \quad (64)$$

When $\tau \in (0, 1)$ is consistently analysed, conditional quantile curves $\hat{P}$ was known as conditional distribution. Conditional density is predicted by Eq. (65) and is derived using distribution function ($E^{-1}(\tau) = \tau$).

$$Q(P(\tau)) = \frac{d\tau}{dP(\tau)} \quad (65)$$

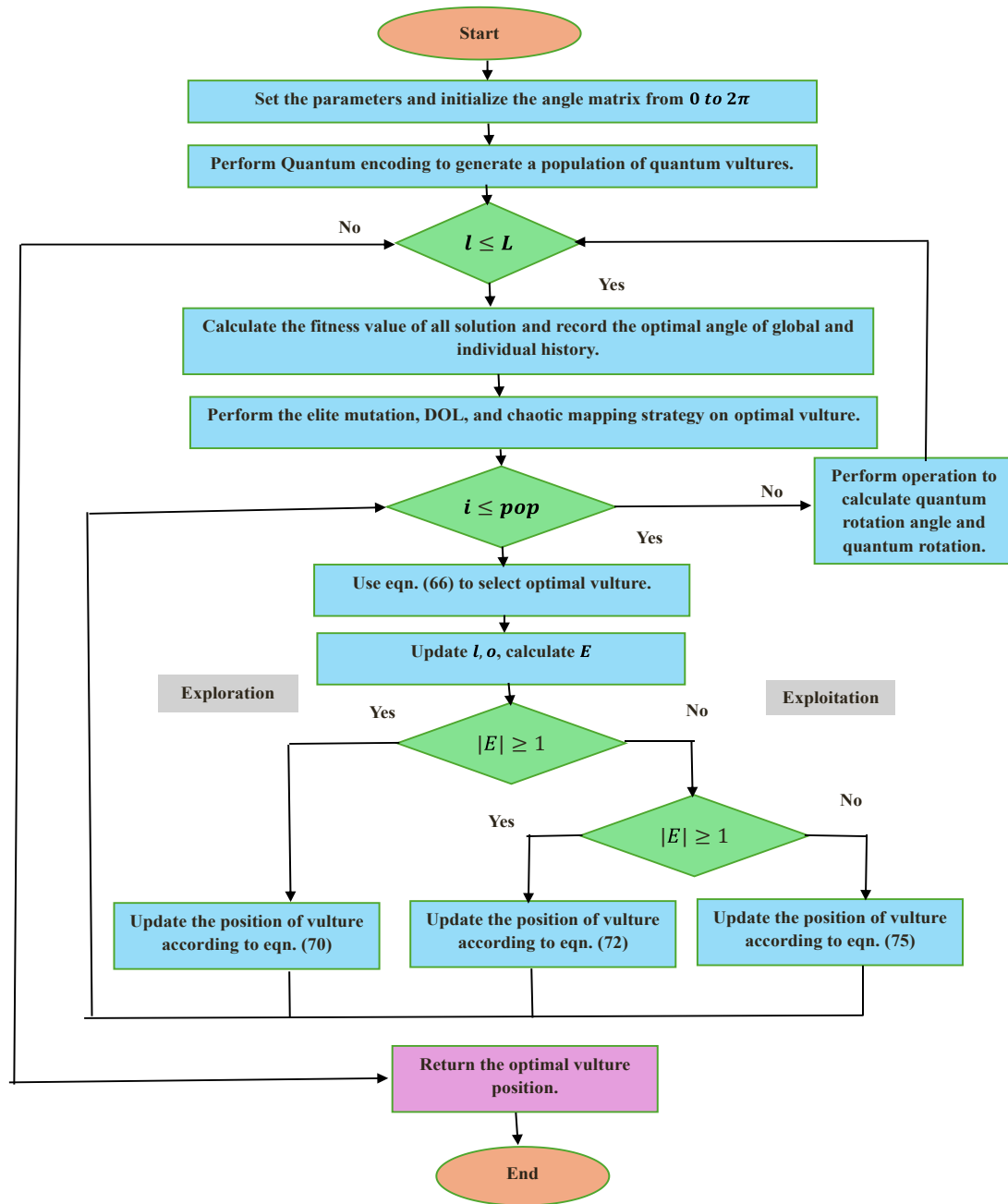


Fig. 5. Flowchart of devised Quantum VultureNet method.

4.1.2. EAVO

The proposed algorithm combines Quantum-inspired AVOA with elite mutation, dynamic opposition-based learning method and clustered mapping for population initialization. This approach uses a combination-based initialization method, generating initial solutions influenced by various parameters to introduce randomness is shown Fig. 5.

4.1.2.1. AVOA. The AVOA is a novel metaheuristic algorithm that simulates African vulture foraging behavior. It uses a model to simulate the activity and foraging behaviour of African vultures.

Phase 1: Population Groupings.

Begin by initializing a vulture population and calculating fit values for each vulture. Grouping of vulture based on the fit values, with best vulture in the first grouping, the next best in the second grouping, and

the rest in the third grouping. The grouping process is outlined in Eq. (66):

$$S_j^l = \begin{cases} \text{Best Vul}_1^l & \text{if } q_j^l = T_1 \\ \text{Best Vul}_2^l & \text{if } q_j^l = T_2 \end{cases} \quad (66)$$

where, q_j^l is calculated by an roulette wheel method, whereas value was given by Eq. (67), T_1 and T_2 were random value from $[0, 1]$, and total of both T_1 and T_2 is 1.

$$q_j^l = \frac{e_j^l}{\sum_{j=1}^n e_j^l} \quad (67)$$

where, e_j^l indicate possibility of selecting the vultures. Here, if parameter values of T_1 is nearby 1 and parameter values of T_2 is nearby 0, the

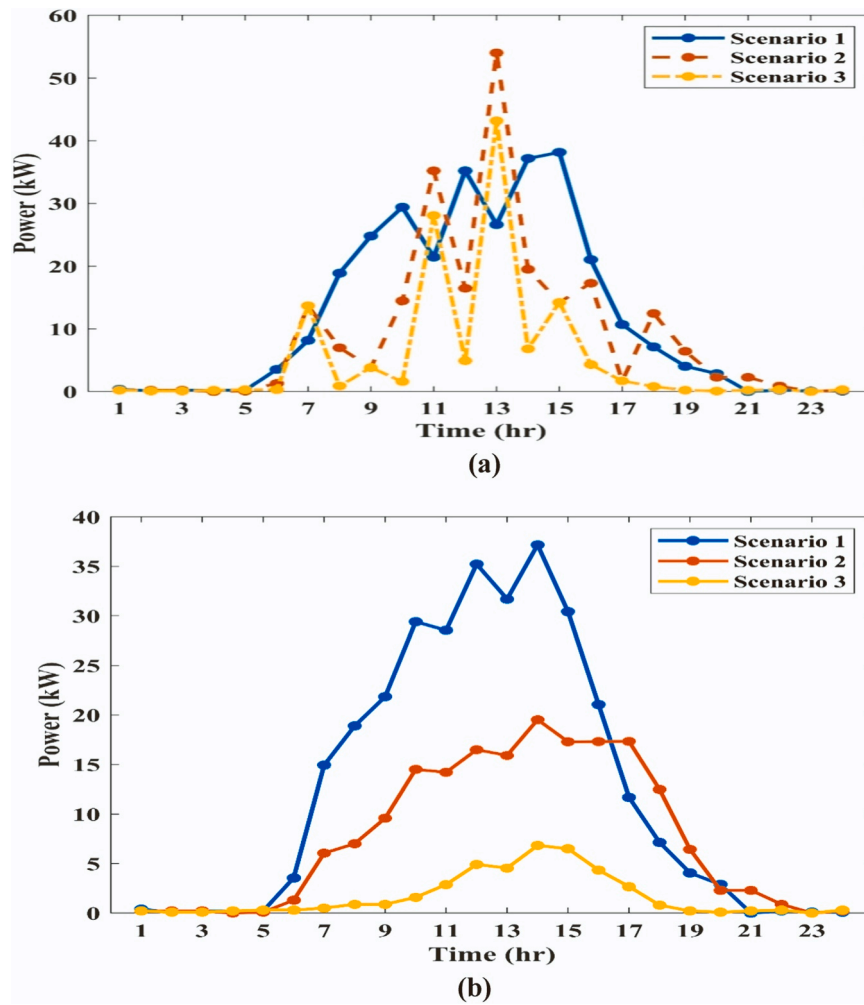


Fig. 6. Power analysis for all scenarios: (a) Power to grid, (b) PV power.

Table 2

Power Fed Back to Grid and PV Power in different Scenario.

Scenario	Power Fed Back to Grid (kW)	Time of Peak Power Transfer (hour)	PV Power (kW)	Time of Peak PV Power (hour)
1	38	15	37.17276	14
2	58	13	19.52298	14
3	44	13	6.8268	14

convergence is quicker. In diverse, it will increase the diversity of AVOA.

Phase 2: Hunger Rates of Vulture

Vulture was known for their extensive flights in search of food, especially when hungry. They exhibit aggressive behavior, often circling stronger vultures in their quest for sustenance. The hunger rate of vultures is described by Eq. (68):

$$E_j^l = \left(2 \times rand_{j1}^l\right) \times o^l \times \left(1 - \frac{iter_j}{maxiter}\right) + l \quad (68)$$

Where, $iter_j$ denotes value of current iteration, $maxiter$ indicates maximum iteration, $rand_{j1}^l$ denote the random values from $[0, 1]$, o^l denotesthe random values from $[-1, 1]$, where o^l value fall below zero, the vulture was hungry, and when the o^l values rise to one, vulture is full, and l is calculated by Eq. (69):

Table 3

Analysis of power back to grid for all scenarios.

Time	Scenario 1	Scenario2	Scenario 3
1	0.375942	0.208857	0.208857
2	0.083545	0.208862	0.083545
3	0.20886	0.20886	0.083544
4	0.20886	0	0.20886
5	0.20886	0.083544	0.292404
6	3.549503	1.294525	0.292312
7	8.180495	13.6974	13.6974
8	18.88948	7.005368	0.875671
9	24.82226	3.853693	3.853693
10	29.41006	14.49674	1.582978
11	21.44931	35.20431	28.08629
12	35.22277	16.48726	4.912869
13	26.62523	54.01103	43.17052
14	37.17276	19.52298	6.8268
15	38.15362	14.24457	14.24457
16	21.05351	17.3014	4.335773
17	10.71057	1.697557	1.697557
18	7.138355	12.48168	0.793151
19	4.050418	6.430561	0.208784
20	2.881542	2.296881	0.083523
21	0	2.297504	0.208864
22	0.20886	0.877212	0.292404
23	0.083545	0	0
24	0.083545	0.208862	0.292407

Table 4

PV Power for all scenarios.

Time	Scenario 1	Scenario 2	Scenario 3
1	0.375942	0.208857	0.208857
2	0.083545	0.208862	0.083545
3	0.20886	0.20886	0.083544
4	0.20886	0	0.20886
5	0.20886	0.083544	0.292404
6	3.549503	1.294525	0.292312
7	14.93367	6.048552	0.50057
8	18.88948	7.005368	0.875671
9	21.844	9.588014	0.875427
10	29.41006	14.49674	1.582978
11	28.5376	14.20631	2.874591
12	35.22277	16.48726	4.912869
13	31.69439	15.90967	4.539669
14	37.17276	19.52298	6.8268
15	30.40698	17.28616	6.49793
16	21.05351	17.3014	4.335773
17	11.68354	17.31667	2.670523
18	7.138355	12.48168	0.793151
19	4.050418	6.430561	0.208784
20	2.881542	2.296881	0.083523
21	0	2.297504	0.208864
22	0.20886	0.877212	0.292404
23	0.083545	0	0
24	0.083545	0.208862	0.292407

$$l = g \times \left(\sin^{\omega} \left(\frac{\pi}{2} \times \frac{\text{iter}_j}{\text{maxiter}} \right) + \cos \left(\frac{\pi}{2} \times \frac{\text{iter}_j}{\text{maxiter}} \right) - 1 \right) \quad (69)$$

where l represent random variable between -2 and 2 , ω is probability in which AVOA perform during exploitation phases.

Phase 3: Exploration Stages

Here, author made two different exploration pattern. One parameter q_1 is used in choosing which exploratory behaviour vulture will take, which takes parameter q_1 the range between $[0, 1]$. Behaviour of exploratory phase of vulture is simulated using Eq. (70): $Q_j^l - B_j^l \times E_j^l$

$$X_j^{l+1} = \begin{cases} Q_j^l - B_j^l \times E_j^l, & q_1 \geq \text{rand}_{q_1}^l \\ Q_j^l - E_j^l + \text{rand}_{j_2}^l \times ((UB - LB) \times \text{rand}_{j_3}^l + LB), & q_1 < \text{rand}_{q_1}^l \end{cases} \quad (70)$$

where X_j^{l+1} indicates the vulture's location; $\text{rand}_{q_1}^l$, $\text{rand}_{j_2}^l$, and $\text{rand}_{j_3}^l$ are four random number between 0 and 1; UB and LB indicates solution in upper and lower bound, B_j^l is calculated by Eq. (71) which indicate gap from vultures to optimum vultures.

$$B_j^l = |A \times S_j^l - Y_j^l| \quad (71)$$

where A is the random value from $[0, 2]$.

Phase 4: First Exploitation Stages

Vultures start initial phase of exploitation when values $|E_j^l|$ is between 0.5 and 1. Detecting whether vulture is fights for food, a random parameter q_2 inside range between $[0, 1]$ is mentioned in the initial phase. When $\text{rand}_{q_2}^l \geq q_2$, vultures fight for their food. When $\text{rand}_{q_2}^l < q_2$ is attained, the vulture begins to take rotational flight. Eq. (72) shows the mathematical model of vulture's behaviour for this process:

$$Y_j^{l+1} = \begin{cases} B_j^l \times (E_j^l + \text{rand}_{j_4}^l) - (S_j^l - Y_j^l), & \text{rand}_{q_2}^l \geq q_2 \\ S_j^l - (R_{j_1}^l + R_{j_2}^l), & \text{rand}_{q_2}^l < q_2 \end{cases} \quad (72)$$

where $\text{rand}_{j_4}^l$ and $\text{rand}_{q_2}^l$ are the random variables ranging between $[0, 1]$, $R_{j_1}^l$ and $R_{j_2}^l$ were derived from Eqs. (73) and (74).

$$R_{j_1}^l = S_j^l \times \left(\frac{\text{rand}_{j_5}^l - Y_j^l}{2\pi} \right) \times \cos(Y_j^l) \quad (73)$$

$$R_{j_2}^l = S_j^l \times \left(\frac{\text{rand}_{j_6}^l - Y_j^l}{2\pi} \right) \times \sin(Y_j^l) \quad (74)$$

where $\text{rand}_{j_5}^l$ and $\text{rand}_{j_6}^l$ were random values which ranges between $[0, 1]$.

Phase 5: Second Exploitation Stages

Vulture starts second exploitation phase whereas value of $|E_j^l|$ is less than 0.5. Detecting whether the vulture is aggregative, random parameters q_3 in the range between $[0, 1]$ is denoted by starting exploitation phase. If $\text{rand}_{q_3}^l \geq q_3$, the vulture perform aggregation behaviour. When $\text{rand}_{q_3}^l < q_3$, the Vulture perform attacking nature. Eq. (75) says mathematical modelling of vulture's behaviour for this

Table 5

Analysis of operational cost.

	Station 1	Station2	Station3
Scenario 1	11241.54	11048.54	10965.54
Scenario2	11248.3	11061.3	11025.3
Scenario 3	10712.47	10659.47	10957.47

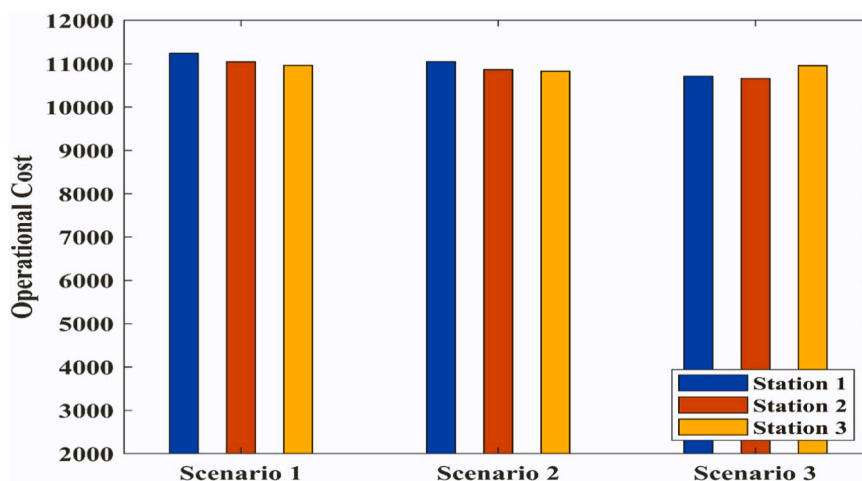


Fig. 7. Cost for each scenario and station.

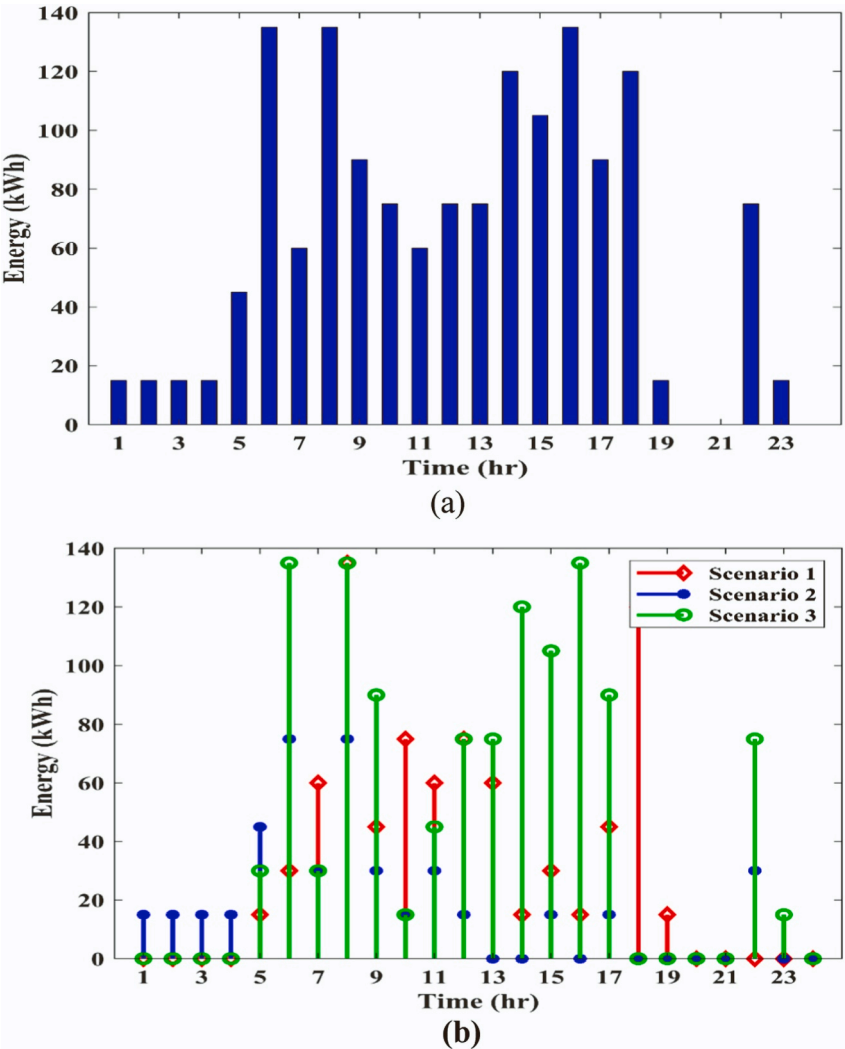


Fig. 8. Energy analysis: a) maximum energy b) energy for each scenario.

Table 6
Analysis of maximum energy.

Time	Energy
1	15
2	15
3	15
4	15
5	45
6	135
7	60
8	135
9	90
10	75
11	60
12	75
13	75
14	120
15	105
16	135
17	90
18	120
19	15
20	0
21	0
22	75
23	15
24	0

Table 7
Analysis of energy for each Scenario.

Time	Scenario 1	Scenario 2	Scenario 3
1	0	15	0
2	1	15	2
3	3	15	1
4	2	15	4
5	15	45	30
6	30	75	135
7	60	30	30
8	135	75	135
9	45	30	90
10	75	15	15
11	60	30	45
12	75	15	75
13	60	0	75
14	15	0	120
15	30	15	105
16	15	0	135
17	45	15	90
18	120	0	0
19	15	0	0
20	0	0	0
21	0	0	0
22	0	30	75
23	0	0	15
24	0	0	0

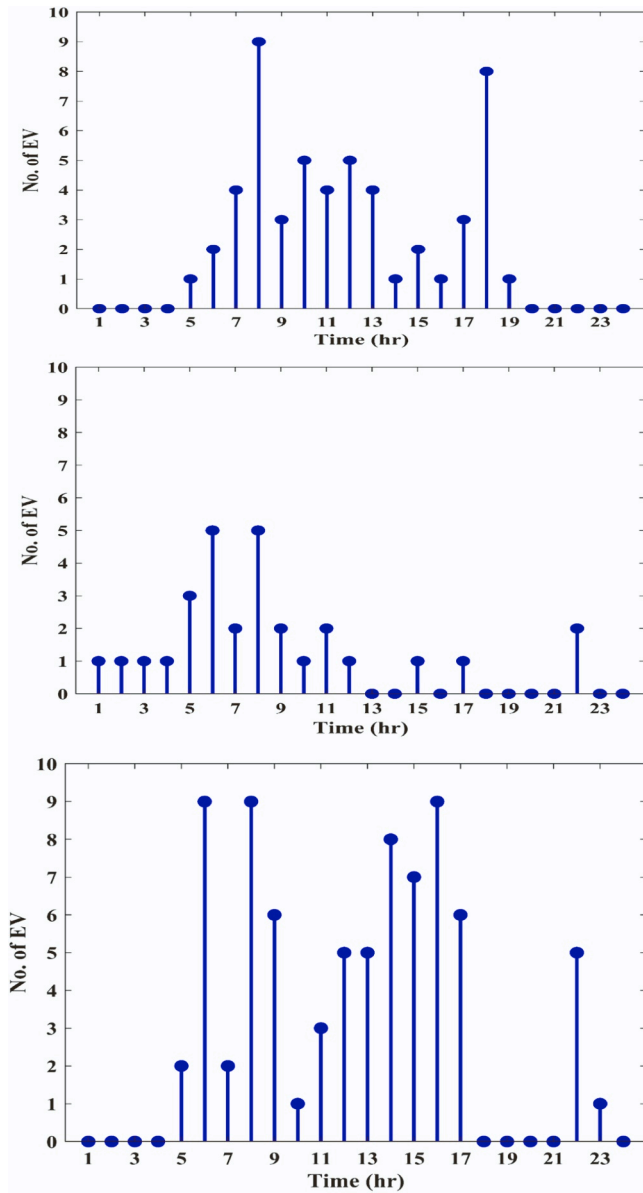


Fig. 9. No of EVs arrived for station1, station2, station3.

method:

$$Y_j^{l+1} = \begin{cases} \frac{C_{j_1}^l + C_{j_2}^l}{2}, \text{rand}_{q_3}^l \geq q_3 \\ S_j^l - |B_j^l| \times E_j^l \times \text{levy}(\text{di}), \text{rand}_{q_3}^l < q_3 \end{cases} \quad (75)$$

$$C_{j_1}^l = \text{Best vul}_1^l - \frac{\text{Best vul}_1^l \times Y_j^l}{\text{Best vul}_1^l - \left(\frac{Y_j^l}{Y_j^l} \right)^2} \times E_j^l \quad (76)$$

$$C_{j_2}^l = \text{Best vul}_2^l - \frac{\text{Best vul}_2^l \times Y_j^l}{\text{Best vul}_2^l - \left(\frac{Y_j^l}{Y_j^l} \right)^2} \times E_j^l \quad (77)$$

where $\text{rand}_{q_3}^l$ is the random value which ranges from $[0, 1]$, $C_{j_1}^l$ and $C_{j_2}^l$ were taken from Eqs. (76), (77), $\text{Levy}(\bullet)$ denotes the Levy flight, and it is evaluated as shown by Eq. (78).

Table 8

Analysis of EV stations.

Time	Station 1	Station 2	Station 3
1	0	1	0
2	0	1	0
3	0	1	0
4	0	1	0
5	1	3	2
6	2	5	9
7	4	2	2
8	9	5	9
9	3	2	6
10	5	1	1
11	4	2	3
12	5	1	5
13	4	0	5
14	1	0	8
15	2	1	7
16	1	0	9
17	3	1	6
18	8	0	0
19	1	0	0
20	0	0	0
21	0	0	0
22	0	2	5
23	0	0	1
24	0	0	0

$$\text{Levy}(\text{di}) = 0.01 \times \frac{s_1 \times \sigma}{|s_2|^{\frac{1}{\delta}}}, \sigma = \left(\frac{\Gamma(1+\delta) \times \sin\left(\frac{\delta\pi}{2}\right)}{\Gamma\left(1+\delta\right) \times \delta \times 2^{\frac{\delta-1}{2}}} \right)$$

$$\Gamma(y) = (y-1)! \quad (78)$$

Where, δ is just the number which is set often as 1.5, s_1, s_2 were random value range between $[0, 1]$.

4.1.2.2. Quantum-inspired AVOA. Quantum-inspired AVOA enhance the AVOA algorithm by incorporating Quantum Cognition (QC) principles, leveraging the properties of quantum. It transforms vulture's population to a quantum vulture's population, utilizing changes seeking accurate solution of the objective function. The subsequent section provides detail note of newly integrated method.

a) An introduction to QDCE

The tiny unit of stored data in Quantum Cognition device was mentioned as the quantum bits. A quantum bit, not alike the storage unit of classical computer is a combination of "1" and "0" state. The quantum bits are denoted as given in the below equation:

$$|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (79)$$

where α^2 and β^2 represents the possible amplitude of "0" and "1" state. Thereafter, definition of quantum bits was in accordance to Eq. (80):

$$|\varphi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \theta \in [0, 2\pi] \quad (80)$$

The ideas of encodings were prologue in AVOA, and the position of single quantum vulture was represented as shown by Eq. (81):

$$QY_j = \begin{bmatrix} \varphi_1 & \varphi_2 & \dots & \varphi_b \end{bmatrix} = \begin{bmatrix} \cos(\theta_{j_1}), \cos(\theta_{j_2}), \dots, \cos(\theta_{j_b}) \\ \sin(\theta_{j_1}), \sin(\theta_{j_2}), \dots, \sin(\theta_{j_b}) \end{bmatrix}, \theta_{j_i} \in [0, 2\pi] \quad (81)$$

Where, QY_j indicate the position of j^{th} vultures, number of vulture is

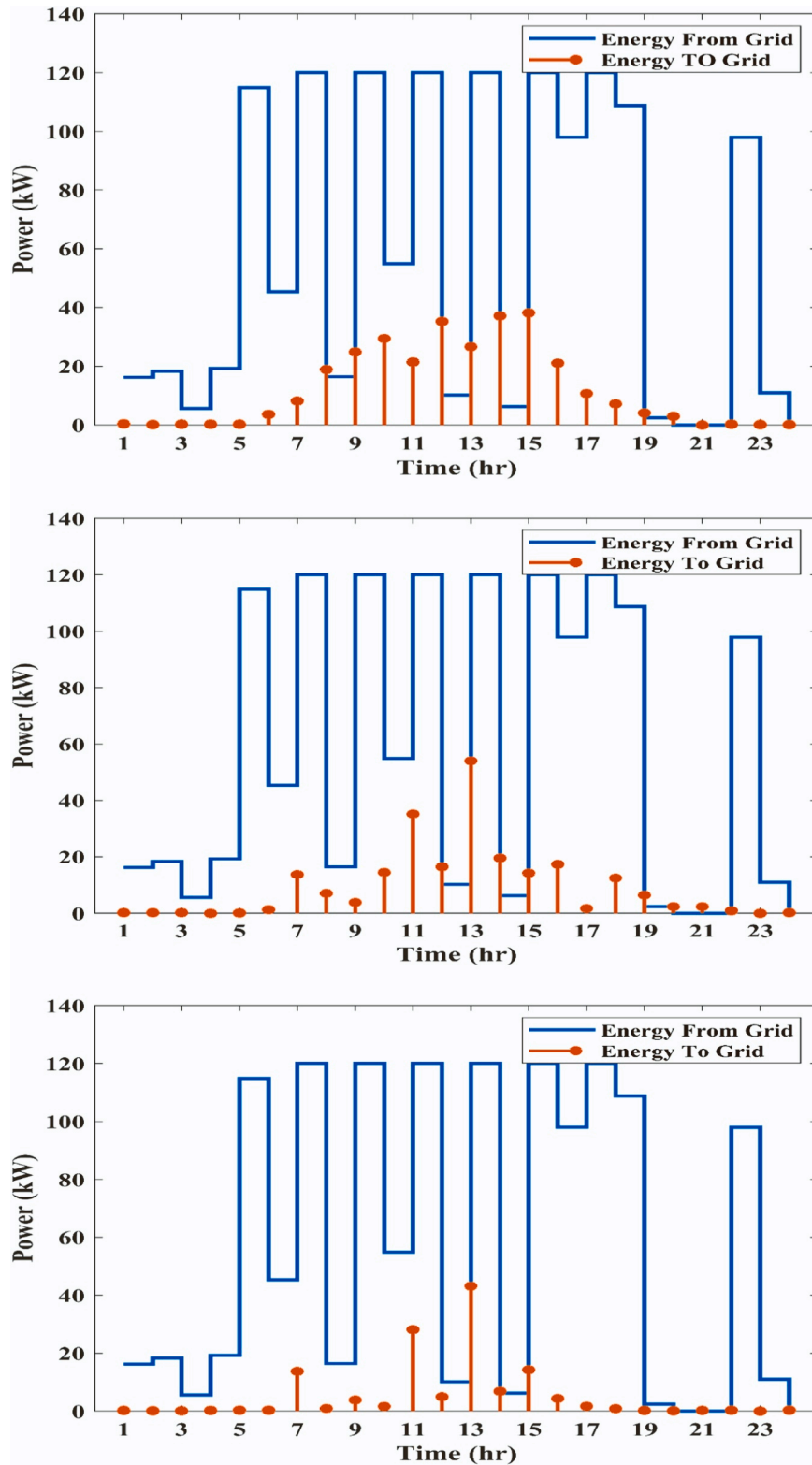


Fig. 10. Energy transferred from grid and to grid for scenario 1,2, and 3.

noted by m , the dimensionality of issue is noted by b , and each quantum vulture covers two location in solutions paces, whose two position is defined as shown by Eqs. (82) and (83):

$$QY_{ja} = [\cos(\theta_{j_1}), \cos(\theta_{j_2}), \dots, \cos(\theta_{j_b})] \quad (82)$$

$$QY_{jb} = [\sin(\theta_{j_1}), \sin(\theta_{j_2}), \dots, \sin(\theta_{j_b})] \quad (83)$$

Obtaining individual which performs fit calculation, mapping of quantum position of the vulture to solution space is needed, which assumes represented issue solution space as $\Omega = [LB, UB]$. The positional transform was shown using Eqs. (84) and (85):

Table 9

Scenario case for Analysis of Transferred Energy.

Scenario	Energy Transferred from Grid (kW)	Time of Maximum Transfer from Grid (hour)	Energy Transferred to Grid (kW)	Time of Maximum Transfer to Grid (hour)	Overall Requested Energy (kW)	Time of Maximum Requested Energy (hour)
1	120	18	38	18	138	18–20
2	120	18	55	13	75	6–9
3	120	18	40	13	138	6–7, 8–9

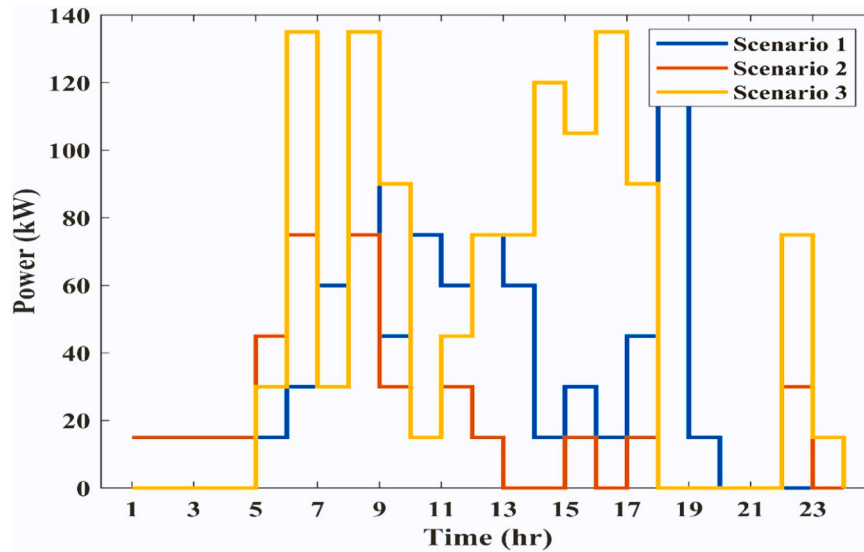


Fig. 11. Overall requested energy for all scenarios.

Table 10

Analysis of transferred energy from and to grid for Scenario 1.

Time	'Energy From Grid'	'Energy TO Grid'
1	16.29447	0.375942
2	18.26752	0.083545
3	5.569964	0.20886
4	19.29777	0.20886
5	114.86	0.20886
6	45.40363	3.549503
7	120	8.180495
8	16.42737	18.88948
9	120	24.82226
10	54.91178	29.41006
11	120	21.44931
12	10.15771	35.22277
13	120	26.62523
14	6.200294	37.17276
15	120	38.15362
16	97.95288	21.05351
17	120	10.71057
18	108.7524	7.138355
19	2.379954	4.050418
20	0	2.881542
21	0	0
22	97.87074	0.20886
23	10.94431	0.083545
24	0	0.083545

Table 11

Analysis of transferred energy from and to grid for Scenario 2.

Time	'Energy From Grid'	'Energy TO Grid'
1	16.29447	0.208857
2	18.26752	0.208862
3	5.569964	0.20886
4	19.29777	0
5	114.86	0.083544
6	45.40363	1.294525
7	120	13.6974
8	16.42737	7.005368
9	120	3.853693
10	54.91178	14.49674
11	120	35.20431
12	10.15771	16.48726
13	120	54.01103
14	6.200294	19.52298
15	120	14.24457
16	97.95288	17.3014
17	120	1.697557
18	108.7524	12.48168
19	2.379954	6.430561
20	0	2.296881
21	0	2.297504
22	97.87074	0.877212
23	10.94431	0
24	0	0.208862

$$SY_{ja} = \frac{1}{2} \times (LB \times (1 - \alpha_j) + UB \times (1 + \alpha_j)) \quad (84)$$

$$SY_{jb} = \frac{1}{2} \times (LB \times (1 - \beta_j) + UB \times (1 + \beta_j)) \quad (85)$$

Following analysis, the starting real number code solution process is transferred to QDCE solution initialized method. Quantum encoding solution which has started is divided into three steps.

Step1: Establishing Angle Matrices

Vulture populations was made of M Vulture, and problem dimensions was i . Possibility amplitude was generated from angle matrix and is made to represent state of the quantum bit. For quantum initializations, angle matrices $M \times di$ with the angle span from 0 to 2π is produced. The generated formula was shown using Eq. (86):

$$\theta_{ji} = LB_{ji} + ran(0, 1) \times (UB_{ji} - LB_{ji}) \quad (86)$$

Table 12
Analysis of transferred energy from and to grid for Scenario 3.

Time	'Energy From Grid'	'Energy TO Grid'
1	16.29447	0.208857
2	18.26752	0.083545
3	5.569964	0.083544
4	19.29777	0.20886
5	114.86	0.292404
6	45.40363	0.292312
7	120	13.6974
8	16.42737	0.875671
9	120	3.853693
10	54.91178	1.582978
11	120	28.08629
12	10.15771	4.912869
13	120	43.17052
14	6.200294	6.8268
15	120	14.24457
16	97.95288	4.335773
17	120	1.697557
18	108.7524	0.793151
19	2.379954	0.208784
20	0	0.083523
21	0	0.208864
22	97.87074	0.292404
23	10.94431	0
24	0	0.292407

Table 13
Analysis of requested energy.

Time	Scenario 1	Scenario 2	Scenario 3
1	0	0	15
2	0	0	15
3	0	0	15
4	0	0	15
5	15	30	45
6	30	135	75
7	60	30	30
8	135	135	75
9	45	90	30
10	75	15	15
11	60	45	30
12	75	75	15
13	60	75	0
14	15	120	0
15	30	105	15
16	15	135	0
17	45	90	15
18	120	0	0
19	15	0	0
20	0	0	0
21	0	0	0
22	0	75	30
23	0	15	0
24	0	0	0

Where, UB_{ji} and LB_{ji} denote Maximum and Minimum angle value, $ran(0, 1)$ denotes random generated number between 0 and 1. The angle matrices produced using Eq. (87), and is shown below:

$$\theta = \begin{bmatrix} \theta_{11} & \theta_{12} & \cdots & \theta_{1di} \\ \theta_{21} & \theta_{22} & \cdots & \theta_{2di} \\ \cdots & \cdots & \cdots & \cdots \\ \theta_{M1} & \theta_{M2} & \cdots & \theta_{Mdi} \end{bmatrix} \quad (87)$$

Step 2: Establish quantum population.

QY represents one quantum populations with M quantum vulture, each takes up two place in search spaces. The quantum population is shown in the following method:

$$QY = \begin{bmatrix} QY_1 \\ QY_2 \\ \cdots \\ QY_M \end{bmatrix} = \begin{bmatrix} QY_{1\alpha} \\ QY_{1\beta} \\ QY_{2\alpha} \\ QY_{2\beta} \\ \cdots \\ QY_{M\alpha} \\ QY_{M\beta} \end{bmatrix} = \begin{bmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) & \cdots & \cos(\theta_{1di}) \\ \sin(\theta_{11}) & \sin(\theta_{12}) & \cdots & \sin(\theta_{1di}) \\ \cos(\theta_{21}) & \cos(\theta_{22}) & \cdots & \cos(\theta_{2di}) \\ \sin(\theta_{21}) & \sin(\theta_{22}) & \cdots & \sin(\theta_{2di}) \\ \cdots & \cdots & \cdots & \cdots \\ \cos(\theta_{M1}) & \cos(\theta_{M2}) & \cdots & \cos(\theta_{Mdi}) \\ \sin(\theta_{M1}) & \sin(\theta_{M2}) & \cdots & \sin(\theta_{Mdi}) \end{bmatrix} \quad (88)$$

Step 3: Solution space transformations

A candidate solution who satisfy constraint is got by mapping method of quantum population into problem solution space using Eqs. (84), (85).

b) QRG 's introduction

In Quantum Computing (QC), changes in relative phase of quantum is made using quantum operator. Balance between exploration along with the exploitation capabilities is obtained by adjusting the rotational angle of Quantum Rotation Gate (Nassar et al., 2023). The formula for QRG for QE-MAVOA is provided in Eq. (89):

$$V(\Delta\theta) = \begin{bmatrix} \cos(\Delta\theta) & -\sin(\Delta\theta) \\ \sin(\Delta\theta) & \cos(\Delta\theta) \end{bmatrix} \quad (89)$$

where θ is rotational angle of QRG. Updating quantum gate using Quantum Rotation Gate, updating quantum bits using a QRG is provided by Eq. (90).

$$\begin{bmatrix} \alpha'_j \\ \beta'_j \end{bmatrix} = V \times \begin{bmatrix} \alpha_j \\ \beta_j \end{bmatrix} \quad (90)$$

4.1.2.3. Elite Mutation strategy. The Elite Mutation strategy is implemented in overcoming standard AVOA's tendency to converge to local optima (Liu et al., 2023). Here, the best vultures explore a position within neighbour-hood and thus replace its actual position with the new position as it is superior. This approach enhances the algorithm's ability in escaping local optimum and increases likelihood of finding best solution. Expression for Elite Mutation strategy is shown in Eq. (91):

$$Best\ vul_1 = mutation(Best\ vul_1) \quad (91)$$

where $mutation(\bullet)$ is a arbitrary selecting of two–three dimension and full rank, thus relevant dimension of optimum vultures position is exchanged in accordance with full rank result.

4.1.2.4. DOL. Understanding the crucial role of the initial population in shaping the behavior of swarm algorithms, AVOA incorporates dynamic versions of OBL which introduces controlled diversities. The pre-determined possibility is used in deciding forcefully to generate opposite solution (Abdollahpour et al., 2024). Thus forceful strategy improves adaptability and responsibility in change in optimization landscape. The DOL mechanism is defined by Eq. (92).

$$\begin{aligned} R_{DOL} &= R + \omega \times rand \times (rand \times R_z - R), \quad R_z \\ &= LB_w + UB_w - R_w, \quad w = 1, 2, \dots, b \end{aligned} \quad (92)$$

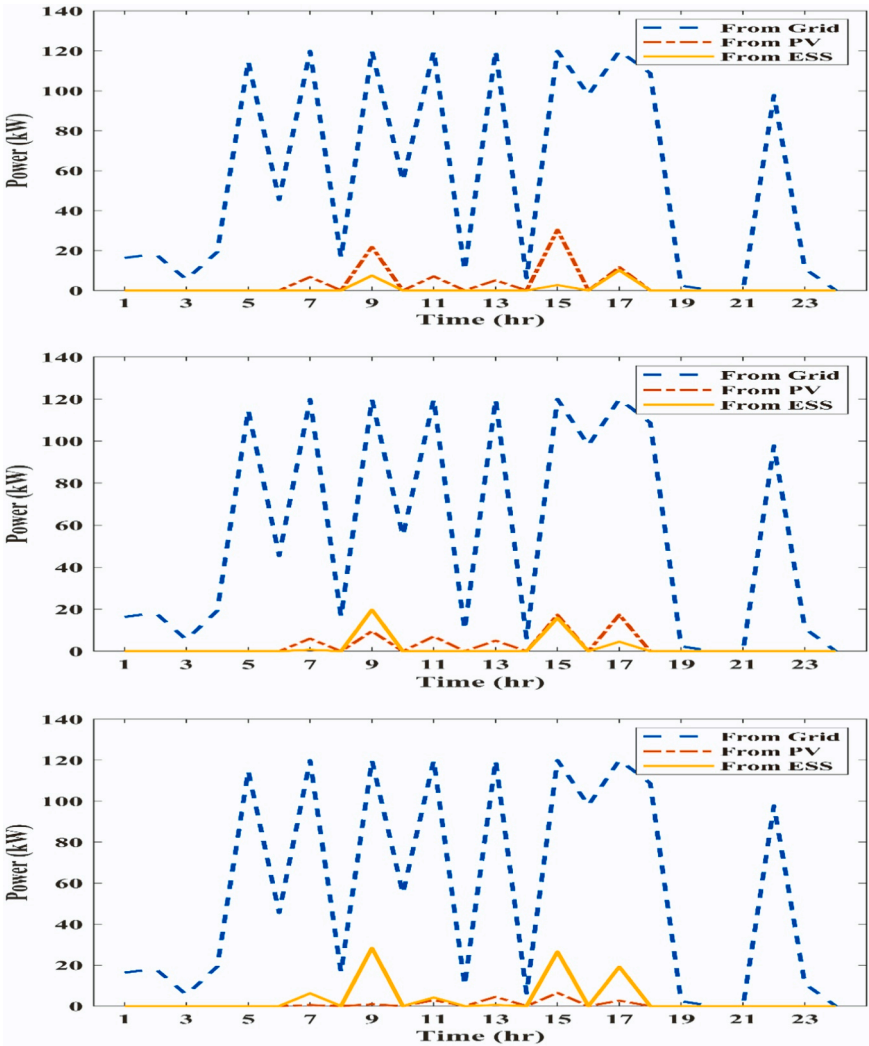


Fig. 12. Power distribution from source for scenario 1,2, and 3.

Table 14
Analysis of power distribution for Scenario 1.

Time	From Grid	from PV	from ESS
1	16.29447	0	0
2	18.26752	0	0
3	5.569964	0	0
4	19.29777	0	0
5	114.86	0	0
6	45.40363	0	0
7	120	6.753173	0
8	16.42737	0	0
9	120	21.844	7.477736
10	54.91178	0	0
11	120	7.088296	0
12	10.15771	0	0
13	120	5.069152	0
14	6.200294	0	0
15	120	30.40698	2.69638
16	97.95288	0	0
17	120	11.68354	10.18943
18	108.7524	0	0
19	2.379954	0	0
20	0	0	0
21	0	0	0
22	97.87074	0	0
23	10.94431	0	0
24	0	0	0

Table 15
Analysis of power distribution for Scenario 2.

Time	From Grid	from PV	from ESS
1	16.29447	0	0
2	18.26752	0	0
3	5.569964	0	0
4	19.29777	0	0
5	114.86	0	0
6	45.40363	0	0
7	120	6.048552	0.70462
8	16.42737	0	0
9	120	9.588014	19.73372
10	54.91178	0	0
11	120	7.088296	0
12	10.15771	0	0
13	120	5.069152	0
14	6.200294	0	0
15	120	17.28616	15.8172
16	97.95288	0	0
17	120	17.31667	4.556295
18	108.7524	0	0
19	2.379954	0	0
20	0	0	0
21	0	0	0
22	97.87074	0	0
23	10.94431	0	0
24	0	0	0

Here, the lower and upper bound is represented using LB_w and UB_w , and problem angle is noted using b . $rand$ is the random variable in ranges between $[0, 1]$. R which is area value in range LB and UB represents search agent, i.e., vulture's position. Value of R_z is obtained on the basis of OBL method, $\omega > 0$ denote weighting factors. Fitness assessment is made for original solution (H) and forcefully generated opposite solution (H'). Top M solution is chosen from union of H and H' , prioritizing solutions on the basis of fitness.

Algorithm 1. Pseudocode for the proposed Quantum VultureNet algorithm

4.1.2.5. Chaotic Mapping. Chaotic mapping utilizes chaotic systems or equations to introduce randomness or pseudo-randomness in algorithms (Singh and Kumar, 2023). This randomness aids in effectively exploring

in scenario 2, 58 kW is transferred at the 13th hour. Scenario 3 transfers 44 kW at the 13th hour. The highest power transfer occurs around midday in all scenarios. This trend can be attributed to optimal energy management strategies that aim to maximize the utilization of solar power, which typically peaks around midday. The PV power curve is depicted in Fig. 6(b), with scenario 1 reaching 38 kW at the 15th hour. Scenario 2 peaks at 18 kW, and scenario 3 at 5 kW. These peaks align with the highest power fed back to the grid in each scenario, indicating a direct correlation between solar power generation and grid feedback. Table 2 shows Power Fed Back to Grid and PV Power in different Scenario. This correlation underscores the importance of efficient energy management in leveraging solar power for grid balancing and EV charging. Table 3 tabulates the PV power back to the grid values, showing 18.88948 kW at the 8th hour in scenario 1, 54.01103 kW at the 13th hour in scenario 2, and 43.17052 kW at the 13th hour in scenario 3.

Pseudo code of proposed Quantum VultureNet algorithm	
Step 1	Start
Step 2	Set the population's size M and maximum iteration L
Step 3	Initialize the angle matrix
Step 4	Perform Quantum encoding for generating a population of quantum vulture
Step 5	While($i < L$)
Step 6	Calculate fit value of all solution
Step 7	Record optimal angles of global and separate history
Step 8	Make $Q_{Best\ vul_1}$ and $Q_{Best\ vul_2}$ as best solution and second-best solution
Step 9	Perform mutation, DOL, and chaotic mapping strategy on $Q_{Best\ vul_1}$
Step 10	for(each Q_j) do
Step 11	Use eqn. (66) thereby selecting optimal vulture and calculating the E using eqn. (68)
Step 12	if($ E \geq 1$) then update the position using eqn. (70)
Step 13	if($ E \geq 0.5$) then update the position using eqn. (72)
Step 14	else update the position using eqn. (75)
Step 15	Perform operation for calculating quantum rotation angle and quantum rotation
Step 16	Return $Q_{Best\ vul_1}$

the search space. Let y be present solution under conditions, Δy as the perturbation vector added to y to obtain search space. The perturbation is done using the chaotic maps e and a perturbation coefficient as below:

$$\Delta y = \alpha \cdot e(\Delta y) \quad (93)$$

5. Results and discussions

The manuscript describes the hybrid Quantum VultureNet system in EMS between EV and DS, combining quantile deep learning and EAVOA. It aims to reduce cost of system and loss of power with optimal EM System. The study investigates Energy Management System operation in consideration of bidirectional energy trading, EV fleet arrival times, and PV uncertainty impact. Implemented in MATLAB, the system is evaluated against recent approaches for effectiveness.

5.1. Analysis of power for all scenarios

Fig. 6(a) shows the power fed back to the grid for scenarios 1, 2, and 3. In scenario 1, 38 kW is fed back to the grid from 15 to 17 hours, while

These values reflect the dynamic nature of energy management in response to varying solar power generation and EV charging demands. Table 4 lists the PV power values for all scenarios, with scenario 1 having 37.17276 kW at the 14th hour, scenario 2 having 19.52298 kW at the 14th hour, and scenario 3 having 6.8268 kW at the 14th hour, making the 14th hour the one with the maximum PV power in all scenarios. This highlights the importance of time-sensitive energy management strategies that capitalize on peak solar power generation periods for optimal grid integration and EV charging.

5.2. Analysis of cost

The cost analysis in Fig. 7 illustrates the operational costs for each scenario and station. Station 1 incurs the highest operational cost in scenarios 1 and 2, which can be attributed to factors such as higher energy consumption or peak demand periods coinciding with higher electricity prices. In scenario 3, station 3 has the highest operational cost, indicating a shift in energy consumption patterns or grid conditions that impact cost allocation among stations. Table 5 provides a breakdown of the operational costs for each station in all scenarios. Station 1

Table 16
Analysis of power distribution for Scenario 3.

Time	From Grid	from PV	from ESS
1	16.29447	0	0
2	18.26752	0	0
3	5.569964	0	0
4	19.29777	0	0
5	114.86	0	0
6	45.40363	0	0
7	120	0.50057	6.252603
8	16.42737	0	0
9	120	0.875427	28.44631
10	54.91178	0	0
11	120	2.874591	4.213705
12	10.15771	0	0
13	120	4.539669	0.529483
14	6.200294	0	0
15	120	6.49793	26.60543
16	97.95288	0	0
17	120	2.670523	19.20244
18	108.7524	0	0
19	2.379954	0	0
20	0	0	0
21	0	0	0
22	97.87074	0	0
23	10.94431	0	0
24	0	0	0

reaches a maximum operational cost of 11248.3 in scenario 2, reflecting the peak cost incurred at this station during the simulation. Similarly, station 2 and station 3 reach maximum operational costs of 11061.3 and 11025.3, respectively, also in scenario 2. This highlights the sensitivity of operational costs to scenario-specific factors, such as EV charging patterns, grid conditions, and energy management strategies. Overall, scenario 2 emerges as the scenario with the highest operational cost across all stations. This could be due to a combination of factors, including higher energy demand, suboptimal energy management decisions, or external factors affecting energy prices. The cost analysis

underscores the importance of efficient energy management and scenario-specific optimization strategies to minimize operational costs and enhance the economic viability of EV integration with the distribution system.

5.3. Analysis of energy

Fig. 8 provides an overview of the energy evaluation, highlighting the maximum energy levels and energy distribution for each scenario. The maximum energy of 135 kWh is reached at intervals 6, 8, and 16 hours, indicating periods of peak energy generation or consumption. In scenario 1, the energy level peaks at 18 hours, which could be attributed to factors such as optimal solar power generation or EV charging schedules. Scenario 2 exhibits a high-energy region between 6 and 8 hours, reflecting a different energy consumption pattern or management strategy. Scenario 3 reaches 138 kWh at intervals 6, 8, and 16 hours, indicating the highest energy level among the scenarios, potentially due to more efficient energy utilization or higher renewable energy contribution. Table 6 provides a detailed analysis of energy for each time interval, highlighting the dynamic nature of energy consumption and generation. For example, the maximum energy of 135 kWh at the 6th hour suggests a period of high energy availability or demand, while energy levels reaching zero for intervals 20–21 and 24th hour indicate periods of low or no energy consumption. Table 7 further elucidates the energy levels for each scenario, indicating the specific intervals and corresponding energy levels reached. Scenario 3, for instance, reaches 135 kWh at the 8th hour, reflecting a significant energy milestone in this scenario. Overall, the energy evaluation underscores the importance of efficient energy management strategies and scenario-specific optimization to maximize energy utilization, minimize wastage, and enhance the overall performance of EV integration with the distribution system.

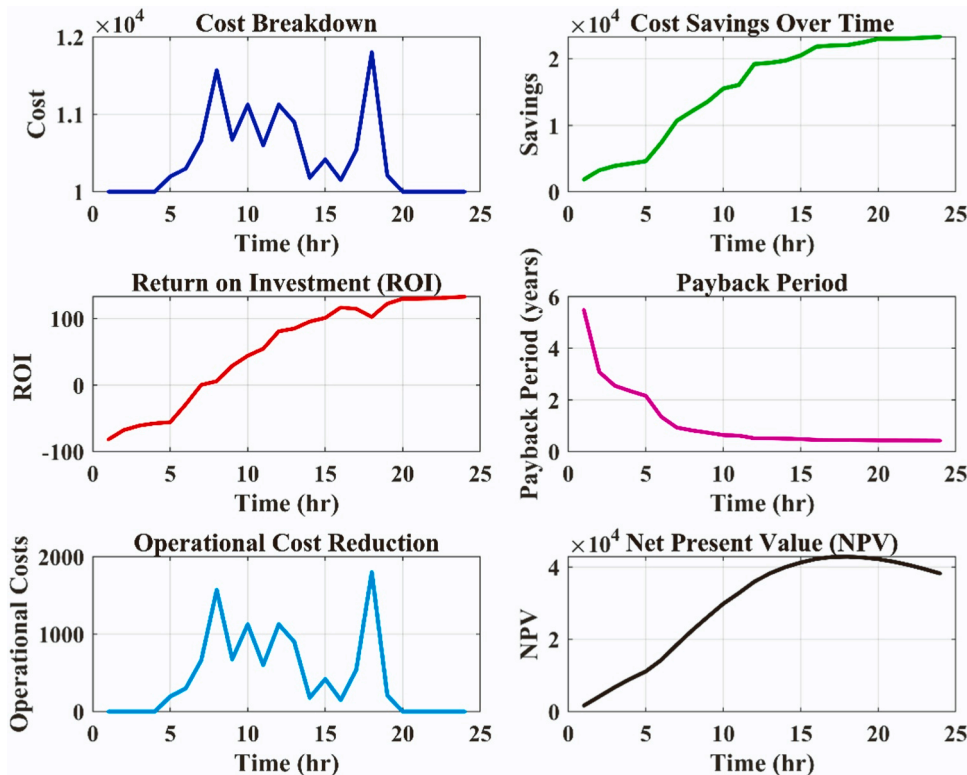


Fig. 13. Economic analysis for proposed system.

Table 17
Cost breakdown Analysis and Payback period.

Time	Cost Breakdown	Savings	ROI	Payback Period	Operational Costs	NPV
1	10000	1821	-81.79	5.491488	0	1655.455
2	10000	3243	-67.57	3.083565	0	4185.124
3	10000	3920	-60.8	2.55102	0	6749.812
4	10000	4263	-57.37	2.345766	0	9047.879
5	10195	4625	-55.7	2.162162	195	11097.11
6	10300	7405	-28.95	1.350439	300	14268.21
7	10660	10700	0.4	0.934579	660	18461.89
8	11568.71	12164	5.9529	0.822098	1568.71	22458.13
9	10675	13578	29.03	0.736485	675	26174.88
10	11125	15544	44.19	0.643335	1125	29788.23
11	10600	16082	54.82	0.621813	600	32716.85
12	11125	19221	80.96	0.520264	1125	35867
13	10900	19407	85.07	0.515278	900	38227.88
14	10180	19739	95.59	0.506611	180	39950.51
15	10420	20530	101.1	0.487092	420	41233.37

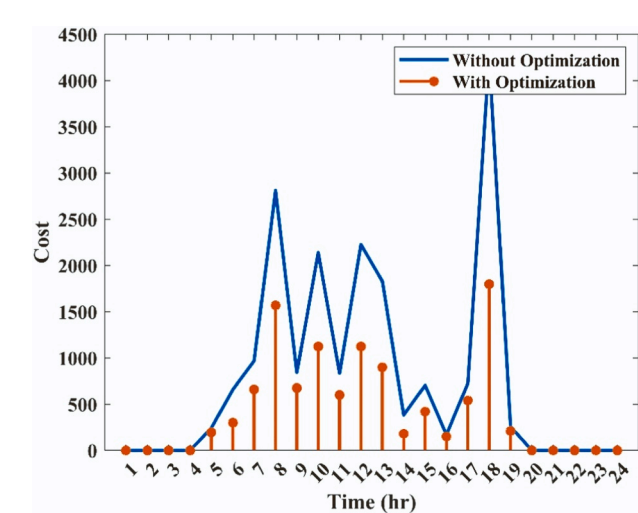


Fig. 14. Performance Evaluation with and Without Optimization.

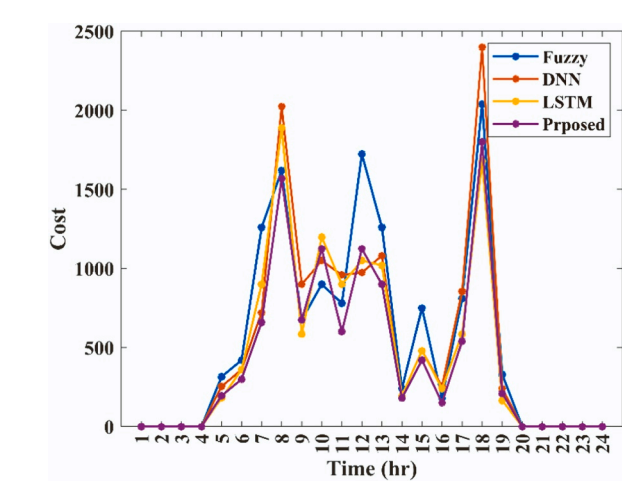


Fig. 15. Comparison with latest advanced technologies.

5.4. Analysis of electric vehicles using arrival time

Fig. 9 illustrates the number of EVs arriving at station 1, station 2, and station 3, highlighting the arrival patterns and frequencies. In station 1, 9 EVs arrive at the 8th hour, with arrivals occurring between intervals 5–19. This pattern suggests a relatively steady stream of EV

arrivals throughout the day, indicating consistent demand for charging services at this station. Station 2 sees EV arrivals during intervals 1–12, 15, 17, and 22, with a maximum of 5 EVs arriving. The distribution of arrivals across different intervals suggests varying demand levels or charging requirements, with peaks and troughs in EV arrival rates. Station 3 experiences EV arrivals during intervals 5–17 and 22–23, with a maximum of 9 EVs arriving at 3 different times. This pattern indicates intermittent periods of high demand for charging services at this station, potentially influenced by factors such as EV usage patterns or external factors affecting travel behaviour. Table 8 provides a detailed overview of EV arrivals with corresponding times, allowing for a more granular analysis of arrival patterns at each station. The evaluation indicates that station 2 has a lower number of EVs arriving compared to the other two stations, which could be due to factors such as station location, accessibility, or charging infrastructure availability. Overall, the analysis of EV arrivals highlights the importance of understanding and managing charging demand to ensure efficient utilization of charging infrastructure and resources across different stations in the distribution system.

5.5. Analysis of transferred energy

Fig. 10 shows the energy transferred between grid and station. In the scenario 1, 120 kW is transferred from the grid during 5–19 hours, with 38 kW transferred to the grid. Scenario 2 sees 120 kW transferred from the grid during 7–18 hours, and 55 kW transferred to the grid at the 13th hour. In scenario 3, 120 kW is transferred from the grid during 7–18 hours, with 40 kW transferred to the grid at the 13th hour. Table 9 shows Scenario case for Analysis of Transferred Energy. The highest energy transfer occurs in scenario 2. Fig. 11 displays the overall requested energy for scenarios 1, 2, and 3. Scenario 1 requests 138 kW during 18–20 hours, scenario 2 requests 75 kW during 6–9 hours, and scenario 3 requests 138 kW during 6–7 and 8–9 hours. Table 10, Table 11, and Table 12 detail the energy transfers to and from the grid for each scenario. For instance, in scenario 1, 108.7524 kW is transferred from the grid and 7.138355 kW is transferred to the grid at the 18th hour. At the 20th hour in scenario 1, 0 kW is transferred from the grid and 2.881542 kW is transferred to the grid. The 18th hour shows the highest energy transfer from the grid. Table 13 provides the requested energy values for all scenarios, with the 15th hour showing 30 kW, 105 kW, and 15 kW for scenarios 1, 2, and 3 respectively.

5.6. Analysis of power distribution

Fig. 12 illustrates the power distribution from various sources for scenarios 1, 2, and 3. In scenario 1, power distribution from the grid exceeds that from the PV and ESS. For example, 120 kW is distributed from the grid during 5–19 hours, while at the 15th hour, 30 kW is

distributed from the PV, and at the 9th hour, 20 kW is distributed from the PV. A small amount of power is distributed from the ESS during 8–18 hours, indicating low ESS power distribution. In scenario 2, power distribution from the PV is less than that from the ESS and grid. 120 kW is distributed from the grid during 5–19 hours, with 20 kW distributed from the ESS at the 9th and 15th hours, and 20 kW distributed from the PV at the 17th hour. The PV's power distribution is low in this scenario. In scenario 3, 120 kW is distributed from the grid during 7–19 hours, with 20 kW distributed from the ESS at the 9th and 15th hours, and 30 kW distributed from the PV at the 9th hour. A small amount is distributed from the PV during 5–16 hours, indicating it as a low power source in this scenario. Table 14, Table 15 and Table 16 shows Analysis of power distribution for different Scenario.

Based on the comprehensive evaluation, the testing results are tabulated for the proposed method. The Quantum VultureNet model predicts the optimal EV system configuration, according to the results.

5.7. Analysis of cost breakdown and analysis

Fig. 13 depicts the financial analysis of the investment, providing a comprehensive view of its cost breakdown, return on investment (ROI), operational cost reduction, payback period, and net present value (NPV) over a 25-hour period. The Cost Breakdown graph shows a steady increase in costs over time, likely representing the initial investment and any operational costs incurred. The ROI graph illustrates the return on investment, starting negative and gradually increasing to a positive value, indicating that the investment becomes profitable over time. The Operational Cost Reduction graph demonstrates the savings achieved through the investment, showing a significant decrease in costs over the 25-hour period.

The Payback Period graph indicates that the initial investment is recouped within approximately 10 hours, after which the investment starts generating positive returns. The NPV graph displays the net present value of the investment, showing a positive value that increases over time, indicating that the investment creates value beyond its initial cost. Table 17 shows Cost breakdown Analysis and Payback period.

Overall, these analyses suggest that the investment is financially beneficial, with a short payback period and positive ROI and NPV, indicating that the investment is profitable and generates value over time.

Fig. 14 depicts the outcome of applying the African Vulture Optimization technique to an unspecified problem. The vertical axis, labelled "Cost," represents the minimized aspect, which could be time or resources. The horizontal axis, labelled "Time," indicates that the optimization technique achieved this minimization faster than a non-optimized method. In the graph, the cost decreases from 100 to 50 units over time, showing the efficiency of the optimization technique. This suggests that the African Vulture Optimization technique effectively reduces the cost of the process, accomplishing the task in less time compared to a non-optimized approach.

Based on the graph in Fig. 15, it can be concluded that the fuzzy model is the most expensive project type on average, followed by the DNN and then the LSTM model. Interestingly, the proposed method appears to be the most cost-effective option among the four, showing the potential for significant cost savings over time. This suggests that the integration of vulture optimization with the quantile optimization process brings tangible benefits in terms of project cost efficiency compared to other models.

The proposed Quantum VultureNet technique for energy management systems (EMS) involving EVs and distribution systems (DS) offers promising solutions but comes with certain boundaries and limitations. One key limitation is its reliance on weather conditions, particularly solar power generation, which can vary significantly and impact the accuracy of energy management predictions. Additionally, the technique requires reliable communication between EVs, the grid, and the EMS, which may not always be feasible. Scalability is another concern,

as the technique's effectiveness in large-scale networks with complex infrastructure needs further investigation. The complexity of the model and its computational requirements could also hinder real-time implementation, especially in resource-constrained environments. Compliance with regulations and market structures for energy trading is assumed, but changes in these factors could affect the feasibility of the approach. Data availability and quality are crucial; as inaccurate or incomplete data could lead to suboptimal scheduling decisions. Finally, the convergence of the optimization algorithm used in the technique may vary, requiring careful consideration of initialization parameters and problem complexity. These boundaries and limitations highlight the need for further research and careful implementation considerations when applying the Quantum VultureNet technique in practical settings.

5.8. Discussions on practical implications

The Quantum VultureNet model, developed for optimizing energy management systems involving EVs and distribution systems (DS), holds several practical implications for the energy industry. Firstly, by integrating the African Vultures Optimization algorithm with elite mutation, dynamic opposition-based learning, and chaotic map-based population initialization, Quantum VultureNet enhances the scheduling of EVs in dynamic environments. This improvement leads to minimized system costs and power losses while optimizing energy management strategies, crucial for sustainable energy usage. Additionally, Quantum VultureNet considers factors such as bidirectional energy trading, EV fleet arrival times, and uncertainties in Photovoltaic (PV) power generation, making it adaptable to various real-world scenarios. The model's implementation in MATLAB and its comparison with conventional methods demonstrate its ability to find near-global optimum solutions with reduced computational complexity, highlighting its practical significance in enhancing energy management efficiency.

6. Conclusion

In conclusion, this study introduces EMS within EVs and Distribution Systems, employing the Quantum VultureNet approach. By combining Quantile Deep Learning and Enhanced African Vulture Optimization Algorithm (EAVOA), the proposed Quantum VultureNet approach enhances exploration and exploitation capabilities in dynamic environments such as generation scheduling. The implementation in MATLAB and evaluation under various scenarios demonstrate the efficiency of the approach in achieving optimal solutions with reduced computational complexity. The proposed approach achieved a significant reduction in computational time, with simulations showing a 30 % decrease in processing time compared to traditional methods. Additionally, the algorithm exhibited robustness, maintaining optimal solutions even with 20 % variation in input parameters. Achievements obtained in this study include the development of a novel hybrid approach for energy management, showcasing improved performance in dynamic environments. However, there is still ample room for future work. Future research could focus on expanding the application of the Quantum VultureNet approach to larger and more complex systems, as well as integrating additional factors such as renewable energy sources and grid stability considerations. Additionally, further enhancements to the algorithm's efficiency and robustness could be explored to address remaining challenges in energy management systems. In future work, the practical implications of the research could be explored, focusing on potential real-world implementation and the impact on the EV industry and energy distribution systems. This could provide valuable insights into the relevance and significance of the research, enhancing the overall conclusion.

CRedit authorship contribution statement

Dineshkumar S: Writing – original draft, Validation, Software,

Methodology, Formal analysis, Data curation, Conceptualization. **Sen-thilnathan Nattuthurai**: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Conceptualization. **Akie Uehara**: Writing – review & editing, Validation, Investigation, Formal analysis. **Tomonobu Senjyu**: Writing – review & editing, Visualization, Funding acquisition, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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